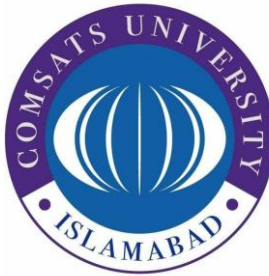


Enhancing Instructor-Focused Usability in Synchronous Online Learning Environments: A Human-Computer Interaction Framework and Prototype Evaluation



MS Thesis

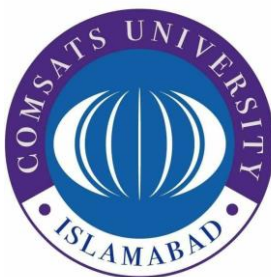
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Spring 2026



Enhancing Instructor-Focused Usability in Synchronous Online Learning Environments: A Human-Computer Interaction Framework and Prototype Evaluation

A thesis Submitted to
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In partial fulfillment
of the requirement for the degree of

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in
Software Engineering

by
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This thesis is submitted to the Department of Computer Science in partial fulfillment of the requirements for the award of the degree of Master of Science in Computer Science.

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Dedication

*D*edicated to My Beloved Parents, My Supervisor: “Prof. Dr. Uzair Iqbal”.

Throughout my academic journey, They were my constant source of inspiration and encouragement. Their unwavering support, boundless love, and infectious enthusiasm for life have shaped the person I am today.

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Abstract

Enhancing Instructor-Focused Usability in Synchronous Online Learning Environments: A Human-Computer Interaction Framework and Prototype Evaluation

By

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This study examines the usability challenges faced by instructors when using Synchronous Online Learning Platforms (SOLPs) such as Zoom and Google Meet. Although these platforms are widely adopted in modern education, particularly after the COVID-19 pandemic, they are primarily designed for general communication rather than instructional purposes, resulting in usability issues that hinder effective teaching. Grounded in Human-Computer Interaction (HCI) principles, the research identifies key problems including complex navigation, inconsistent interfaces, fragmented tools, and increased cognitive workload. To evaluate these issues, both subjective and objective usability measures were employed, including the System Usability Scale (SUS), NASA Task Load Index (NASA-TLX), task completion time, error rates, and success rates. An instructor-centered prototype interface was developed based on ISO 9241-110 usability principles and Nielsen's heuristics to address these limitations. A within-subjects quasi-experimental design was adopted, where instructors performed teaching tasks using Zoom, Google Meet, and the proposed prototype. The collected data were analysed to compare usability performance, efficiency, and cognitive workload across platforms. The findings indicate that the proposed prototype significantly enhances usability by improving task efficiency, reducing errors, and lowering cognitive load. This study contributes an integrated instructor-focused usability framework that combines usability evaluation with cognitive workload analysis. Overall, the research highlights the importance of designing educational platforms that align with instructors' needs to improve teaching effectiveness in synchronous online learning environments.

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List of Abbreviations

<i>HCI</i>	Human–Computer Interaction
<i>SOLPs</i>	Synchronous Online Learning Platforms
<i>SUS</i>	System Usability Scale
<i>NASA-TLX</i>	NASA Task Load Index
<i>ISO</i>	International Organization for Standardization
<i>CLT</i>	Cognitive Load Theory
<i>WCAG</i>	Web Content Accessibility Guidelines
<i>UI</i>	User Interface
<i>UX</i>	User Experience
<i>ANOVA</i>	Analysis of Variance
<i>AHP</i>	Analytic Hierarchy Process
<i>ML</i>	Machine Learning
<i>NLP</i>	Natural Language Processing
<i>NPS</i>	Net Promoter Score
<i>ASL</i>	American Sign Language
<i>COVID-19</i>	Coronavirus Disease 2019

Chapter 1

Introduction

1.1 Overview

The Human-Computer Interaction (HCI), also referred to as Human-Computer Interface, represents an interdisciplinary field focused on the design of usable interactive systems that are efficient and satisfy users' needs [1], [2]. HCI combines ideas from several scientific fields such as computer science, psychology, ergonomics, and design, to improve human-technology interaction [1], [3]. In today's digital environment, usability represents one of the most important factors for determining software-system success, having a direct impact on users' performance and satisfaction from the system [3],[4]. Systems that are hard to use and understand often struggle in obtaining broad acceptance, despite all their technical features.

In educational settings, HCI plays a significant role in designing educational systems and technologies. Educational systems need to handle various complicated tasks of communicating and collaborating between individuals and managing information and processes [1], [5]. Such activities interfere with workflow, generate extra cognitive load, making it harder for educators to concentrate [5], [6]. Therefore, poor usability may be even more harmful for educators.

In the last ten years, there has been tremendous growth in Synchronous Online Learning Platforms (SOLPs) due to improvements in communication technology and increased internet availability. The growth has been enhanced further by the outbreak of the coronavirus disease 2019 (COVID-19), which forced educational establishments around the world to switch from traditional classroom education to online learning. Therefore, Synchronous Online Learning Platforms (SOLPs) became the main method of conducting classes, discussion, and interactions with students [7]. In addition, the change has become permanent and will be a part of modern education going forward, including online learning as an inseparable element of the learning process.

Synchronous online learning platforms provide the possibility to interact with each other instantly using features like video calls, screen sharing, chat, and collaboration capabilities. Some of the commonly utilized Synchronous Online Learning Platforms (SOLPs) include

Zoom, Google Meet, Microsoft Teams, Webex, and BigBlueButton, widely used by universities, colleges, and schools for virtual classrooms [8], [9], [10], [11].

These platforms offer various features that facilitate online teaching activities like screensharing, room-breaking, chats, file sharing, polls, and recording features.[12] However, despite all these features, they can be effective only when the instructors have the capability to use them effectively during their lessons [8],[13]. In this regard, there is a need to note that despite having various features, the effectiveness of their usage may depend on how easy the instructor accesses and uses them.

While these systems have been used widely in education, they were originally designed for general business purposes not for educational purposes [14], [15]. Consequently, there are various challenges related to interface design since these systems have not taken into consideration the actual needs of the instructors in using these interfaces in teaching environments. This mismatch in the design creates usability problems that affect teaching process[16], [17].

There is various usability problems associated with these systems as indicated by empirical literature such as navigation problems, inconsistent layout problems, inability to locate the critical controls, as well as fragmented teaching tools across different menus [13],[16], [18], [19], [20]. These usability problems interrupt workflow and create increased cognitive load on the part of the instructor.

The past SUS-based studies show that many of the existing synchronous platforms have only average levels of usability [4], [13], [21]. Scores below 70 on the scale denote significant problems encountered by users when working with the system [8], [13]. The results emphasize the importance of improving interface designs from a usability perspective.

Besides the SUS rating, subjective measures of workload performed through the NASA Task Load Index (NASA-TLX) show higher mental workload experienced by instructors when carrying out teaching tasks in synchronous settings [22], [23], [24]. A higher workload could lead to stress, inefficiency, and a greater possibility of making mistakes in synchronous online classes. Since teaching is generally associated with complex brain activity, poor design interfaces make this task even harder.[25]

Finally, the objective performance measure should be considered another crucial aspect of usability evaluation. Poorly designed interfaces increase the time needed for completing tasks

and the error rate in synchronous lessons [23], [26].

To solve these issues, HCI research stress the importance of following the usability standards and HCI design principles that are already known and used. Standards like ISO 9241-110 define structured interaction principles that can be used in designing usable systems[2]. The principles offered by Nielsen in his usability heuristics can also be used to improve user interfaces[3]. All these principles focus on key aspects such as system visibility, consistency, avoidance of errors, and learnability.

If these principles are not followed, users might experience confusion, higher cognitive load, and loss of efficiency [2], [3]. Therefore, improving the usability of Synchronous Online Learning Platforms (SOLPs) requires a systematic approach, which will involve using usability standards, instructor's feedback, and evaluation methods.[27]

The study presented aims to solve these problems and to bridge the gap between current practices of software development and instructor's needs by focusing on usability improvements and their impact on teaching performance. It will develop an instructor-centered HCI model and conduct an evaluation of a prototype of an interface designed to enhance usability in Synchronous Online Learning Platforms (SOLPs).

1.2 Motivation

The motivation behind this research arises out of the increasing reliance on synchronous learning environments within modern educational systems as well as the usability problems experienced by instructors during their teaching sessions. When using synchronous learning environments like Zoom and Google Meet, instructors must simultaneously manage multiple activities including giving lectures, moderating classroom discussions, monitoring students' attention spans, answering queries from students, and manipulating other functions on the software interface.

However, most instructors have reported usability problems such as having trouble finding certain controls, having to navigate through complex menus, and managing several tools at once during live sessions. Such usability problems not only disrupt the teaching process but also increase the cognitive burden placed upon instructors while conducting live classes.

From an HCI perspective, systems should be designed to fit user activities and their cognitive abilities. Educators need a user interface that will help them manage classes efficiently without having to create many complexities. The tools available currently focus on communicative

aspects rather than education itself.

The problems associated with poorly designed interfaces, unclear labels, and inconsistency in layout design have been demonstrated both empirically and by educators themselves. This might result in errors which would include wrong configuration settings, missed interactions, and interruption of live lessons.

Thus, there is a need for a new system which will focus on meeting instructors' needs. Using the techniques of creating user-friendly applications and relying on instructors' advice, it is possible to create a better system.

1.3 Problem Statement

Conferencing systems, such as Microsoft Teams[13], Google Meet [9], Webex [10], BigBlueButton [11], and zoom[8],[15] are utilized as basic means of conducting distance learning. These tools enable students and instructors to communicate in real time and collaborate in group activities, [8], [13] thereby simulating the environment of a traditional classroom[9]. However, usability problems can be observed. Previous studies have reported several usability issues in synchronous learning platforms, including complex navigation patterns experienced by 58% of users [13], cluttered user interfaces reported by 46% of users [13], inconsistent experiences across web, desktop, and mobile platforms [13], fragmented toolsets [18], and unnecessarily complex interface designs[12], [15], [16], [17]. The real-life experiences of teachers in classrooms also reveal problems with the use of menus and class controls, which are difficult to see or find. These problems contribute to low usability scores (<70), like the SUS[8], [13], decrease performance [19], [28], increase cognitive load, and directly affect task efficiency as well as error rates in live sessions [11], [13], [22]. Addressing these usability problems may improve the effectiveness, efficiency, and overall satisfaction of instructors using online learning platforms. It can provide higher results in terms of SUS, lower mental workload according to NASA-TLX, and lower error rates. Thus, usability should be considered indispensable for effective distance learning [5], [13], [18], [19], [20], [28], [29] .

1.4 Research Questions

The study is formulated about the following research questions:

RQ1: What usability problems arise in the use of synchronous online learning platforms (for example, Zoom and Google Meet) by instructors?

RQ2: Which usability principles and guidelines are violated in existing systems?

RQ3: Is it possible to achieve higher usability results by means of designing an instructor-

centered prototype?

1.5 Research Objectives

The main objectives of this research paper are:

1. Identification of usability problems experienced by instructors in the context of synchronous online learning platforms;
2. Analysis of violations of usability guidelines using ISO 9241-110 and Nielsen's heuristics;
3. Development of a high-fidelity instructor-centered prototype interface;
4. Evaluation of the usability of the prototype using objective and subjective methods;
5. Compare usability performance of the prototype to that of existing systems

1.6 Research Significance

By addressing the issue of instructor usability in synchronous online learning platforms, the research will provide useful data for the HCI field and educational technology sphere. In particular, the findings will draw conclusions about the usability problems that instructors have and the need to apply usability guidelines to make educational interfaces more user-friendly. Since instructors play a crucial role in organizing teaching process in synchronous conditions, improvements in their interaction with platforms can lead to better results in terms of online education quality. Usability is important for instructors since a user-friendly system allows them to focus on teaching, not platform navigation and other technical aspects.

The instructor-focused design model and prototype presented in this study provide guidance that may help developers and research create an effective online teaching platform capable of providing classroom management functionality and reducing the instructor's cognitive load. Such a focus is reasonable since numerous existing platforms are oriented towards communication without considering the specific requirements of instructors' work at school. [12]The study proposes a focused usability standpoint which will be beneficial for future developments of such platforms.

In addition, the research provides valuable information based on results obtained during usability experiments as well as statistical analysis conducted during the study. Instead of using hypotheses or personal judgments, the author uses usability statistics to make comparisons between the existing platforms and the proposed prototype. In this way, research and educators are provided with solid evidence regarding usability of the proposed solution.

Finally, the value of the study consists in the application of concepts like HCI and workload analysis to the field of online education as well as usability assessment of web-based technologies. As a result, the current study makes a significant contribution not only in terms of a practical prototype but also in terms of the methodology used by research.

1.7 Research Scope

In this research, the usability of synchronous learning applications is evaluated from the point of view of the instructors. Two of the most popular platforms for such purposes, namely Zoom and Google Meet, are analyzed and compared to a recently designed instructor-oriented prototype. The goal of such analysis is to find out whether an application designed for the instructor's convenience is more efficient compared to other applications currently available.

Three fundamental components of usability according to ISO 9241-11 model are effectiveness, efficiency, and satisfaction [2]. The effectiveness dimension measures how accurate and successful instructors' actions are performed by means of using such applications. Efficiency refers to how long it takes to perform the task. Finally, satisfaction describes users' subjective perceptions regarding the software system.

To measure usability of synchronous learning applications comprehensively, the research uses objective and subjective measurements. As for the former one, task completion time and success rate will be used. Subjective measurements will involve SUS questionnaire and NASA-TLX test. For usability analysis, the present study adopts different parameters, allowing performance evaluation as well as the assessment of perceived ease of use and mental workload of the system under investigation.

Instructors, lecturers, and tutors with prior experience in the delivery of online courses through synchronous learning platforms form the sample size. The involvement of participants with relevant experience is very critical to provide an authentic evaluation based on real-life situations.

This study does not conduct a detailed evaluation of user-centered usability but only considers usability from the perspective of the instructors' needs. Furthermore, this study confines itself to synchronous online learning environment only. It does not extend to asynchronous online learning environments as well. This approach has been adopted intentionally to confine the scope of this study to instructor-student interaction and teaching process.

1.8 Thesis Contribution

Contributions of this study include the following:

- Identification and analysis of key **instructor-centered usability challenges** in synchronous online learning platforms
- Development of an **instructor-centered Human–Computer Interaction (HCI) framework** tailored for real-time teaching environments
- Integration of **usability evaluation (effectiveness, efficiency, satisfaction)** with **cognitive workload analysis (NASA-TLX)** into a unified framework
- Design and implementation of a **high-fidelity prototype interface (Figma-based)** focused on improving instructor interaction and task efficiency
- Empirical evaluation of widely used platforms (**Zoom and Google Meet**) using both **objective and subjective usability measures**
- Application of **heuristic evaluation (Nielsen’s usability heuristics)** to identify usability guideline violations in existing systems
- Statistical validation of usability differences using **Repeated Measures ANOVA and post-hoc analysis**

1.9 Thesis Structure

The rest of this paper is organized as follows. In Chapter 2, a review of background literature regarding the use of synchronous online learning platforms is provided, together with an extensive literature review of past research conducted in the field of usability testing and interface design. It highlights possible gaps in current literature and the theoretical and empirical justification for conducting the present study.

Chapter 3 describes the research methodology used in this study, including the conceptual model, the hypothesis formulated, the procedure of creating the prototype, the experimental design, and the data collection and analysis methods. It details the planning and implementation of the experiment and the justification of the methodology used for the purpose of measuring the usability of the product.

Chapter 4 details the findings obtained from the usability test experiments performed using Zoom, Google Meet, and the proposed prototype. It also presents statistical analyses based on the collected data and discusses the implications of the findings in relation to the research questions posed and the objectives of this study.

Finally, Chapter 5 draws conclusions about the key findings of this study and discusses its

limitations and recommendations for further research. It considers the importance of usability design in instructor-centered synchronous online learning platforms and how these findings can be applied in the future.

Chapter 2

Literature Review

2.1 Background and Related work

The synchronous feature of the Synchronous Online Learning Platforms (SOLPs) has been the primary research focus for the years 2019–2025. Software such as Microsoft Teams [13], Google Meet [9], Webex [10], BigBlueButton [11], and Zoom [8], [14] have been examined in detail from the aspect of the interfaces' simplicity/complexity, navigational challenges, and instructor-dominated usability issues [8], [9], [10], [13], [14], [15], [26]. The years surrounding the early pandemic (2019–2021) experienced a significant growth in the adoption of Synchronous Online Learning Platforms (SOLPs). For instance, adoption increased dramatically during the pandemic period [7], [8]. Previous studies revealed issues of managing participation, layout differences, and difficulties locating features in Microsoft Teams, Webex, and Google Meet [8], [22], [26], [30]. These early findings indicate that, although online learning tools are commonly implemented, their design and user interaction do not conform to the known rules of teaching and usability. This suggests that there is a vast difference between technology use and good teaching. After the pandemic, researchers have applied usability tests, such as the System Usability Scale (SUS), to various platforms using surveys [4], [13], [21]. The results of the measurements revealed that 58% of users experienced issues with navigation, and 46% of the respondents considered the layouts messy or confusing [20]. This has led to low SUS scores below 70 because of concealed controls and various designs in web, desktop and mobile applications [11], [22]. Webex menus were found to be clumsy, and Google Meet breakout rooms lacked features. These two problems resulted in increased working time and decreased productivity [5], [19]. Many usability testing and feedback studies were conducted between 2023 and 2024 to validate previous results. An extensive survey with over 5,000 user reviews found that complicated interfaces, confusing navigation, and technical issues caused 56% of negative comments [20], [29]. Usability testing showed that simpler, combined interfaces improved task success rates by 22% and reduced perceived mental effort by 18%, as measured using the NASA-TLX scale [11]. Also, controlled experiments in group learning showed that better interfaces helped learners remember information 15% better, confirming the link between usability and learning results [13], [28]. Recent studies (2024–2025) examined issues teachers have with designing for usability and accessibility. Surveys of hybrid teachers indicated that they are overwhelmed by personal activities in breakout rooms, chat options, and

screen-sharing options [28], [31]. The usability testing of the real-time monitoring dashboards showed that too much information prevented the instructor's quick response [32], [33]. Proctoring and assessment software caused apprehension among students and instructors because aspects of control were hidden [15], [31]. Accessible research has highlighted the importance of flexibility in design, correct captions, and screen-reader support for diverse learning styles [16], [30]. Experiments in rural areas with limited Internet showed that the users needed intuitive and straightforward mobile interfaces [34], [35]. Table 2.1 also presents a comparison of various evaluation measures and approaches adopted from the literature. To better understand existing design approaches, several HCI and usability frameworks are reviewed below.

2.1.1 Existing HCI and Usability Frameworks in Synchronous Online Learning Platforms (SOLPs)

2.1.1.1 FRAMEWORK 1: General HCI Usability Model

Several studies in Human–Computer Interaction adopt generalized usability frameworks based on ISO 9241-11, where system usability is evaluated through three primary dimensions: effectiveness, efficiency, and user satisfaction [2]. These frameworks provide a structured approach for assessing how well users can achieve their goals within a system. In synchronous e-learning environments, this issue acquires significance because teachers need to accomplish many activities without making users dissatisfied.



FIGURE 2.1 Dimensions of Usability According to ISO 9241-11 (Source: [2])

The Figure 2.1 shows a typical Human–Computer Interaction usability model where usability is measured based on effectiveness, efficiency, and user satisfaction. These dimensions provide a basis for evaluating system performance in interactive environments. Although this model forms a solid ground for assessing system performance in terms of usability, it fails to capture the effects of cognitive workloads and interaction complexities in learning environments.

2.1.1.2 FRAMEWORK 2: Cognitive Workload + Interaction Model

Aside from usability dimensions, various research studies have incorporated different cognitive workloads models to evaluate user interactions. For instance, the NASA Task Load Index (NASA-TLX) measures mental effort, mental demand, and frustration levels when doing a task[24]. Under synchronous learning platforms, teachers must juggle many tasks at once, which raises the cognitive load immensely. Therefore, incorporating cognitive workload models into usability frameworks gives a much better evaluation of the systems' performance.

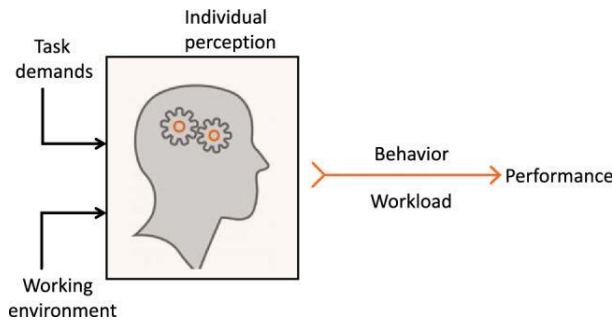


FIGURE 2.2 Cognitive Workload Model in Human–Computer Interaction (Source: [36])

Figure 2.2 shows a framework that illustrates how user interactions with the system create cognitive workload with dimensions of mental demand, effort, and frustration, all of which can be measured by NASA-TLX. However, while cognitive workload models provide some understanding of users' mental effort, they do not consider task performance and usability.

2.1.1.3 FRAMEWORK 3: Instructor-Centered Interaction Framework

Recently, studies have shown the importance of having instructor-centered frameworks in designing synchronous online learning systems. Unlike system-centered frameworks that focus on students, the instructor-centered model is concerned about the user's responsibility of controlling the communication, content, and interactive functions at the same time [32], [33]. These frameworks illustrate the need for interfaces to allow task integration and quick decision-making without increased cognitive workload. However, there are still many platforms that fail to incorporate these aspects into their design, thus making them unusable in live classes.

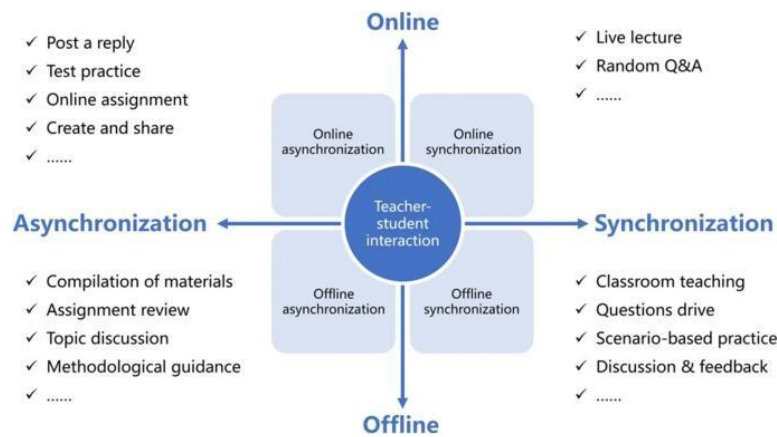


FIGURE 2.3: Instructor-Centered Interaction Framework for Synchronous Online Learning (source: [37])

Figure 2.3 below shows an instructor-centered interaction framework that highlights the link between the tasks of the instructor, the system interface, and student interaction in synchronous online learning. As much as these frameworks focus on the interaction of instructors with the interface, there is little evidence to show usability and workload together.

From the above frameworks, usability in synchronous online learning systems is multidimensional and involves various factors such as task performance, cognitive workload, and interaction. Though past frameworks have made important contributions in usability evaluation and system design, most of the frameworks are either general or focus on specific aspects like workload or accessibility. It is clear from the above discussion that there is a need for an all-inclusive framework that addresses the issues of usability, cognitive workload, and the demands of real-time instruction. It is due to this reason that the framework discussed in this paper has been developed.

These frameworks provide some useful information about the process of usability evaluation. But the above frameworks fail to incorporate the specific aspects of interaction problems faced by instructors, the requirements of real-time instruction, and issues related to cognitive workload into one single model. For this purpose, an instructor-centered usability framework that includes usability factors, cognitive workload, and task-based interaction will be developed in this research.

2.2 Accessibility in Synchronous Online Learning Platforms

Apart from the traditional aspects of usability, accessibility is one of the core factors considered during interface design, particularly when such platforms are used by a range of instructors who have varying needs. The Web Content Accessibility Guidelines provide a solid base for

designing inclusive interfaces. These guidelines focus on improving access to electronic content by persons with disabilities, guaranteeing that people who are visually impaired, hearing impaired, or motorically challenged can interact with the platforms easily. This includes attributes such as:

- **Perceivability:** Text alternatives for non-text content, sufficient contrast, and video captions.
- **Operability:** Navigation via keyboard or assistive technology and avoiding time-based obstacles.
- **Understandability:** Clear, easy-to-understand language and predictability in design.
- **Robustness:** Compatibility with current and future assistive technologies.

Empirical evidence suggests that implementing accessibility will have positive effects not only on the user group but also on the overall performance of the system in general. Importantly, ISO 9241-171 and Section 508, which applies to American-based systems, present solid guidelines for accessible design. The instructor-oriented prototyping process (refer to Section 3.7) interacts with several different standards, such as the font size and contrast adjustment and screen reader support, as outlined in WCAG 2.1.

2.3 Usability Evaluation Methods

Usability testing is considered an essential part of HCI research used to measure the efficiency of user-system interactions. HCI researchers use various methods for usability testing that include subjective and objective tests as well as usability inspection.

2.3.1 System Usability Scale (SUS)

The System Usability Scale (SUS) represents a commonly used tool for measuring system usability. Originally created by Brooke [4], the scale involves ten statements about system usability that users rate using a five-point Likert scale, varying from strongly disagree to strongly agree. The result of applying the SUS is a score between 0 and 100, where the higher the score is, the better the usability of the interactive system perceived by the user is. It should be noted that a score around 70 is considered the minimum acceptable level of usability, while a lower score may indicate the presence of usability problems and difficulties using the system.

2.3.2 NASA Task Load Index (NASA-TLX)

NASA Task Load Index (NASA-TLX) is a reliable subjective method of measuring the cognitive workload associated with a wide range of tasks [22], [23], [24]. Unweighted NASA-

TLX involves six dimensions that include mental demands, physical demands, temporal demands, performance, effort, and frustration. Subjects are required to evaluate each dimension by considering their subjective impressions of performing specific tasks under consideration. In this regard, higher ratings in NASA-TLX mean higher cognitive demands in terms of mental workload when accomplishing the tasks while lower ratings refer to less cognitive demands. NASA-TLX is especially effective when dealing with complex systems containing multiple tasks being performed simultaneously.

2.3.2.1 Cognitive Load Theory (CLT) and Its Relevance to Usability and Mental Workload

The Cognitive Load Theory (CLT) by Sweller (1988) forms the fundamental basis for analyzing mental workloads and cognitive functions within learning exercises.[38] According to Sweller, human cognitive capabilities are limited and must be optimized to facilitate learning and enhance performance. In synchronous learning settings online, CLT allows the consideration of design deficiencies of the interface as factors contributing to extraneous cognitive loads, thus reducing instructors' effectiveness. CLT categorizes cognitive loads into:

1. Intrinsic Load: the difficulty of the content or activity.
2. Extraneous Load: cognitive activities emanating from how the content is presented but not necessarily part of the learning process.
3. Germane Load: the cognitive functions allocated to understanding and interpreting the content.

In this regard, extraneous cognitive load plays a significant role in synchronous online teaching tools such as Zoom and Google Meet because their designs add to instructors' mental demands. By designing systems that minimize extraneous load, the interface can improve the usability and task performance of instructors, as well as reduce the mental workload.

2.3.3 Objective Usability Measures

Besides subjective questionnaire, usability testing may rely on objective criteria. Such criteria are based on measurable indicators associated with user actions within the course of performing tasks and provide objective evidence concerning the quality of system performance. Objective usability testing often involves the following performance metrics:

- Time to perform tasks
- Error rates
- Success rate of completing tasks

Time to perform tasks refers to the amount of time needed by users to perform assigned tasks. In turn, error rates show the number of mistakes committed during task completion while the

success rate of completing tasks implies the percentage of successful completion of a task without mistakes.

2.5 Comparative Analysis of Previous Studies

The topic of usability of synchronous online learning platforms has been extensively discussed in academic literature. Different dimensions of system usability, such as interface, navigation, cognitive load, and user satisfaction, have been studied in this field. The usability of popular synchronous tools such as Microsoft Teams, Zoom, and Google Meet were analyzed using usability questionnaires and observation tools. Studies conducted within the framework of usability evaluations using System Usability Scale have found low usability scores of synchronous learning platforms, which suggests user interface problems with system interfaces [13], [21].

Some experiments using methodology have been carried out to determine user performance when completing certain tasks. The objective usability measures such as the time taken to complete tasks and errors are measured to find usability problems in system interfaces [23], [26]. Findings show that complex interface designs and disorganized control mechanisms take more time and lead to many errors.

User experiences with online learning systems have been studied through examining their cognitive workloads. Various studies carried out on these aspects using the NASA-TLX measure have found that instructors and learners often suffer from heavy mental workloads because of carrying out several tasks at once in online learning environments [22], [23], with a high possibility of lowering user performance.

In addition to usability experiments, various studies have used user feedback and app reviews to determine usability problems in online learning applications. In one of such experiments, users' sentiments were analyzed to determine that most criticisms made were about complexity and difficulties of interfaces and navigation as well as other technical problems in apps [20], [29].

Some other researchers have focused on examining prototypes and interface designs for enhancing usability of online learning applications. These experiments showed that simpler interfaces and use of dashboards improve performance when undertaking tasks in online learning systems [28], [39].

Table 2.1 provides a comparative overview of several studies evaluating the issues of usability

in synchronous online learning platforms using different approaches and criteria.

TABLE 2.1: Comparison of Evaluation Criteria and Methods on the Topic of Usability Issues in Synchronous Online Learning Platforms (SOLPs)

Ref	Platforms	Methodology	Dataset / Participants	Evaluation Measures	Limitations
[5]	Mixed collaborative tools	Empirical experiment	University students	Retention rate	Focus on discussion tools rather than full conferencing platforms; does not evaluate real-time teaching interactions. Our study focuses on full real-time teaching workflows with 44 instructors.
[6]	Mixed educational tools	AHP survey	Students & instructors	Feature ranking	Uses AHP ranking instead of real usability testing; covers general tools, not platform-specific interaction. Our study provides real usability testing with instructors.
[7]	Zoom, Google Meet, and other online media	Surveys and perception analysis	Students and teachers (number not specified)	Ease of use, accessibility, navigation, engagement	Focused on COVID-19 context; may not reflect current usage patterns; perception-based, not task-based evaluation. Our study evaluates current platforms and task-based metrics.
[8]	Zoom	Literature synthesis	—	Thematic analysis	Based entirely on secondary literature; lacks empirical validation. Our research bridges this gap by empirically testing usability with real instructors.
[9]	Zoom, Google Meet	Survey-based study	Students (number not specified)	User satisfaction, interface clarity, engagement	Based only on student perspectives; instructor experience not included; limited

					depth of usability evaluation. Our study addresses this by focusing on instructor usability.
[10]	Zoom, Google Meet, Microsoft Teams (likely)	Usability assessment via surveys and interviews	Students and instructors (number not specified)	Interface design, navigation, accessibility, overall usability	Regional study limits global applicability; sample size not clearly defined. Our study involves a diverse, global sample of instructors.
[11]	Gather Town, Zoom	Experimental user testing	Students (number not specified)	Interface clarity, navigation, engagement	Small-scale experiment limits reliability; comparison limited to two platforms. Our study includes 44 instructors, ensuring a larger sample size and broader platform comparison.
[12]	Discord (Coordination tool)	System design + pilot	Instructors & students	Task completion rate	Focus on Discord limits applicability to mainstream platforms. Our study enhances this by evaluating widely used platforms like Zoom and Google Meet.
[13]	Microsoft Teams	SUS questionnaire	Higher-ed students	SUS score (62.5)	Sample size-related limitations have been considered. While previous research may have faced issues with the small number of participants, the current study with 44 teachers provides richer results regarding the usability problems faced by teachers.
[14]	Zoom, Teams	User testing + cognitive style analysis	Students (number not specified)	Navigation ease, interface clarity,	Cognitive style approach may not represent diverse users; sample size

				engagement, satisfaction	likely limited. Our study evaluates a more diverse instructor sample, accounting for varied teaching experience and technical expertise.
[15]	Zoom, Teams, Google Meet (likely)	Surveys	Students and educators (number not specified)	Usability, interface design, navigation, security	Focus on developing country context limits generalization; security focus reduces emphasis on usability depth. Our study ensures broad applicability by evaluating real teaching scenarios.
[16]	Zoom	Controlled test	University students	Anxiety scale, task errors	Focused only on exam-related scenarios; does not represent full teaching workflows. Our study addresses this by considering full teaching workflows.
[17]	Mixed institutional tools	Case studies + interviews	Administrators & faculty	Workflow analysis	Qualitative case studies lack quantitative usability metrics; focus on workflows rather than interface usability. Our study combines quantitative usability metrics with real-world feedback from instructors.
[18]	Zoom, Teams, Meet	Observational testing	University instructors	Task success rate	Limited to basic conferencing tasks; does not reflect complex teaching workflows. Our study expands on this by assessing real-time teaching workflows, including class management tasks.
[19]	Teams, Zoom	Qualitative	Hybrid-	Thematic	Small qualitative

		analysis	teaching faculty	coding	sample limits generalizability. We extend this by including a larger and more varied instructor sample across different educational settings.
[20]	Zoom, Teams, Google Meet (likely)	Predictive analytics + survey/feedback	Students (number not specified)	Interface layout, engagement, and satisfaction metrics	Predictive model lacks real-world validation; limited cross-platform generalization. Our study fills this gap with empirical usability testing across real-world platforms.
[21]	Zoom, Teams, Webex, Meet	Large-scale UX survey	>1000 global users	SUS, NPS	Large-scale survey lacks task-based usability validation; general user data; not instructor-specific. Our research involves task-based usability with a targeted group of 44 instructors.
[22]	Mixed	Controlled experiment	Students	NASA-TLX cognitive load	Conducted in a simulated/prototype environment, not real platforms. Our research involves 44 instructors using real platforms.
[23]	Teams, Zoom	Task-based usability	Instructors	Response time, error count	Limited participant pool reduces reliability of findings. Our study includes 44 instructors, offering a more robust and diverse sample for assessing usability in real-world teaching environments.
[25]	Mixed	Online education Qualitative analysis of patterns	Multiple online courses	Engagement indicators	Theoretical analysis without empirical validation; no measurable usability metrics.

					Our study provides empirical usability testing to complement the theoretical analysis.
[26]	Virtual communication tools (Zoom, Meet, Teams, etc.)	Experimental and ML-based analysis of UX patterns	Users of virtual communication tools	Navigation ease, interface usability, and engagement metrics	Dataset size and composition not specified; results may vary across different platforms. Our study clearly defines sample size and includes cross-platform comparisons, ensuring reliability and consistency.
[27]	BigBlueButton	Descriptive analysis	—	Feature change log	Descriptive feature analysis without empirical usability testing; no user-based evaluation. Our study includes empirical usability tests involving 44 instructors.
[28]	Conceptual dashboards	Prototype design + evaluation	Higher-ed students	Perceived clarity, decision time	Prototype-based evaluation without real platform deployment. We address this by comparing real platforms alongside the prototype.
[29]	Mixed (user reviews)	NLP & deep learning	Public app-store reviews	Sentiment ratio, topic modelling	Relies on user-generated app reviews, which may be biased or inconsistent. Our study overcomes this by using a controlled, empirical approach with instructors.
[30]	Teams, Zoom	Pilot deployment	Instructors	Setup time, usability score	Small-scale pilot limits statistical validity; short-term evaluation only. While our sample size of 44 instructors is smaller than large-scale studies, it still provides a more

					representative group of real instructors.
[31]	Accessible custom platform	Development + user testing	Students with accessibility needs	Task success, SUS	Custom platform not comparable to mainstream systems; accessibility-focused; lacks general usability evaluation. Our research integrates accessibility features into mainstream platforms.
[32]	Mixed	Collaborative design	Rural secondary schools	Usability rating	Study conducted in a specific rural context; limited generalizability. Our study improves generalizability by involving instructors from various educational settings.
[33]	Zoom, Teams	Quantitative survey	Deaf ASL users	Accessibility preferences	Focused only on accessibility for a specific user group; does not address general usability across all users. Our study addresses this by incorporating general usability concerns alongside accessibility features.
[34]	Mixed	Participatory design	Rural students	Task success	Prototype-based testing; lacks real-world deployment. We bridge this gap by testing prototypes alongside real platforms.
[35]	Zoom (concept adaptation)	Prototype evaluation	Young learners	Engagement metrics	Prototype-based environment limits ecological validity; focus on young learners only. Our study addresses this by evaluating adult instructors in real

					teaching environments.
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2.6 Usability Guidelines Violations

An operational analysis revealed that Zoom and Google Meet platforms violate several usability guidelines from ISO 9241-110 and Nielsen's usability heuristics. These violations were identified through a combination of literature review and guideline analysis. Specific details of these violations, including the related interface elements and how they were addressed in the prototype, are thoroughly discussed in **Chapter 3, Section 3.5** and **Section 3.6**

For example:

- Zoom violates multiple interaction guidelines, such as **5.1.3 Suitability for Task**, **5.5.2 Error Prevention**, and **5.4.1-5.4.3 Consistency**.
- Google Meet violates **5.6.4 Adaptive Behavior** and **5.7.1 Support for Learning**.

Please refer to **Chapter 3** for a comprehensive breakdown of these violations and how they informed the design requirements for the instructor-focused prototype.

2.7 Research Gap

While existing studies have investigated usability issues in synchronous online learning platforms thoroughly, most of them focus on student viewpoints, general usability criteria, or specific aspects of systems. However, there is a paucity of scholarly articles that combine instruction-centered interaction design with the analysis of cognitive workload and actual usability tests of several systems. Furthermore, it is rare to find research studies that suggest a comprehensive model of usability guidelines, workload assessments, and real-time teaching. This research contributes to addressing the identified gaps by developing and validating an instructor-centered usability framework that includes usability criteria, cognitive load, and real-time interaction of teaching.

Chapter 3

Research Methodology

3.1 Introduction

This research aims at overcoming the usability problems that are known from synchronous online learning systems, through a structured method of Human-Computer Interaction (HCI). Using a systematic research approach, it involves the creation of a theoretical framework based on the known usability principles, as well as the development of hypotheses to assess the performance of the system. In terms of research design, it will involve a quantitative experimentation that will compare the usability of current tools like Zoom and Google Meet with a new instructor-oriented prototype. The new system is developed through a systematic design and development process based on the interaction principles defined by ISO 9241-110 and the usability heuristics proposed by Nielsen. Usability assessment uses both objective measures such as time taken for tasks, number of errors made, the percentage of successful accomplishment of tasks, as well as subjective measures like System Usability Scale (SUS) and NASA Task Load Index (NASA-TLX). Finally, collected data is analyzed using suitable statistical tests like repeated measures ANOVA to find out if there are any meaningful differences in terms of efficiency, effectiveness, and user satisfaction. Figure 3.1 presents the overall methodology adopted for this study.

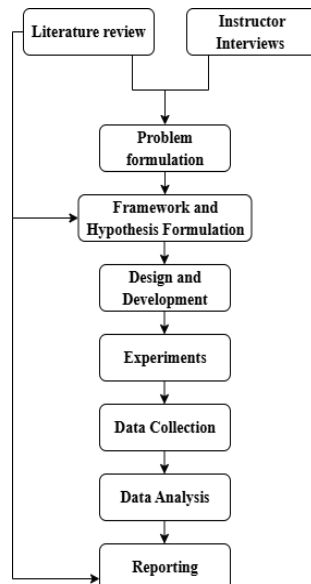


FIGURE 3.1: The process followed for conducting the research.

3.2 Research Design

In essence, the main objective of this study is to explore how usability challenges affect the teaching process of instructors in an online environment and whether redesigning the interface based on Human-Computer Interaction (HCI) can improve their efficiency and satisfaction in the process of teaching. To achieve this goal, the study used within-subjects quasi-experimental design, which is applicable in measuring the performance of usability in terms of numeric factors such as the time taken to complete the task, number of errors made, usability scores, and cognitive workload and where a set of participants interacted with several systems. In this case, instructors are required to execute similar tasks using three different systems, namely Zoom, Google Meet, and the proposed system prototype designed for instructors. The usage of common participants for all systems allows the evaluation of their usability efficiency while eliminating any differences among individual abilities such as instructive skills, technical knowledge, and experience in using the Internet. The current research uses both objective and subjective indicators of usability. Subjective measures include System Usability Scale (SUS) and NASA Task Load Index (NASA-TLX), whereas objective measures include task completion time, error rate, and task success rate [3].

Additionally, the prototype adheres to usability guidelines based on Human-Computer Interaction literature [1], [3] like ISO 9241-110 interaction guidelines and Nielsen's usability heuristics, hence complying with known Human-Computer Interaction guidelines. The data gathered is then subjected to analysis to determine whether the developed prototype results in any significant usability gains compared to current systems.

3.3 Research Framework

The conceptual framework of this research, illustrated in Figure 3.2, is based on the usability model defined in ISO 9241, which evaluates system usability through three key dimensions: effectiveness, efficiency, and user satisfaction [2]. These dimensions form the core usability outcomes of the study and are used to assess how effectively instructors can perform teaching-related tasks using synchronous online learning platforms. In this research, the independent variable is the type of online learning platform used during the experiment. The three systems being analyzed include Zoom, Google Meet, and the suggested instructor-centered prototype.

The dependent variables represent the results of usability, which were measured through the experimental analysis, namely time to complete a task, errors, percentage of successfully completed tasks, cognitive load, and users' satisfaction. Time to complete a task and the number of errors made by users can be considered measures of the system's efficiency and

effectiveness, respectively. Cognitive load and user satisfaction are subjectively measured using validated scales for assessing system performance, such as the NASA Task Load Index (NASA-TLX) and System Usability Scale (SUS). To guarantee the validity and reliability of the experimental data, several control variables are used, namely the similarity of tasks assigned to users, consistent participant samples tested on different systems, equal conditions for testing, and standardized instruction for participants.

The conceptual framework presented in Figure 3.2 assumes that improvements in interface design based on usability principles will lead to better usability outcomes.

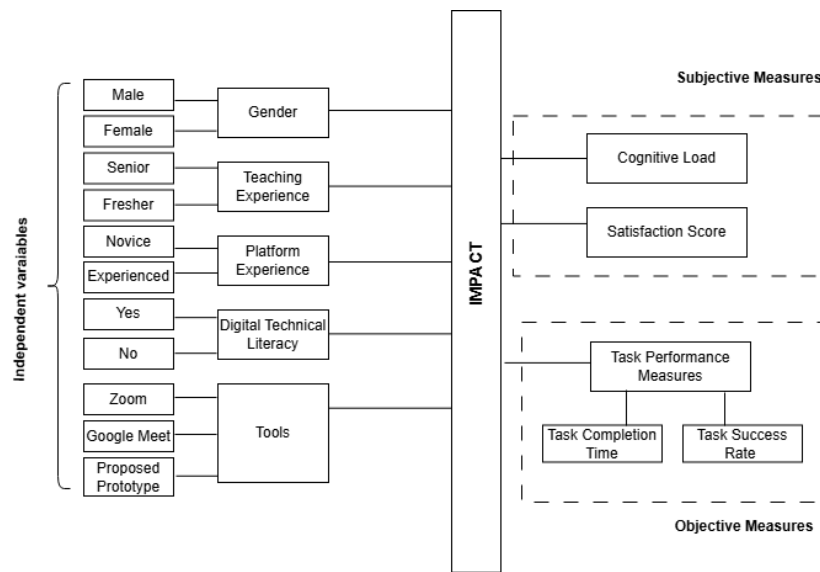


FIGURE 3.2: Proposed conceptual framework for evaluating usability outcomes with different factors

The research framework is grounded in Cognitive Load Theory (CLT), which asserts that cognitive capacity is limited and that extraneous cognitive load such as the mental effort caused by poorly designed interfaces, should be minimized to enhance user performance. Through measuring the cognitive load by using the NASA-TLX measure, this study attempts to investigate the effect of the design of interfaces on cognitive load of the instructors when performing tasks. The aim of this research is to test whether the proposed instructor-oriented prototype reduces the extraneous cognitive load compared with other tools like Zoom or Google Meet.

3.4 Hypothesis Development

Based on research objectives and conceptual framework, four hypotheses were developed to evaluate the usability improvements provided by the proposed prototype.

H1: “There is no significant difference in the effectiveness of task completion between the instructor-centered prototype and existing platforms such as Zoom and Google Meet.”.

H2: “There is no significant difference in the efficiency of task completion between the instructor-centered prototype and existing platforms such as Zoom and Google Meet”

H3: “The proposed prototype will achieve higher usability ratings compared to existing platforms.”

H4: “The proposed prototype will achieve lower cognitive workload compared to existing platforms.”

This set of hypotheses examines the effects of the new interface on task efficiency and effectiveness in comparison with current systems. Furthermore, it evaluates the subjective results of the usability experiment in terms of SUS and NASA-TLX results to investigate if the new interface leads to better user experience and decreased mental effort for teachers.

3.5 Identification of Usability Issues in Existing Platforms

Before developing the prototype, usability issues in existing platforms were identified through a combination of literature review and usability guideline analysis.

An investigation of the literature reveals that there are multiple synchronous learning platforms that violate established usability guidelines. These violations result in greater frustration among users, greater cognitive load, and reduced task efficiency.

TABLE 3.1: Usability Guidelines Violated by Zoom and Google Meet (Based on ISO 9241-110 and Nielsen’s Guidelines)

Tool	ISO 9241-110 Violated Principles	Nielsen’s Usability Heuristics Violated
Zoom	5.1.3, 5.2.2, 5.4.1–5.4.3, 5.5.2–5.5.3, 5.6.2–5.6.4, 5.7.2–5.7.3	1, 3, 5, 6, 8, 9
Google Meet	5.6.4, 5.7.1	1, 7, 9, 10

An **operational analysis** was conducted to identify which usability guidelines were violated by Zoom and Google Meet illustrated in Table 3.1 The analysis considered usability principles from two major sources:

- **ISO 9241-110 interaction principles**
- **Nielsen’s usability heuristics**

The results indicated that Zoom violates several interaction principles represented in Table 3.2 including 5.1.3 task suitability, 5.5.2 error prevention, 5.4.1- 5.4.3 consistency, and 5.7.1 support for learning. Similarly, Table 3.3 represents Google Meet violations related to 5.6.4 adaptive behavior and 5.7.1 support for learning.

These findings informed the design requirements for the proposed instructor-focused prototype.

3.6 Operationalization of Usability Guidelines

To translate usability principles into practical design requirements, an **operational document** was created. This document identifies the usability guidelines violated by existing platforms represented in Table 3.2 and Table 3.3, describes how these issues can be addressed in the prototype.

TABLE 3.2: ISO 9241-110 Usability Principles Violated by Zoom

Clause No	Principle	Definition	Example Implementation in Prototype
5.1.3	Suitability for Task	System should support users in completing tasks efficiently.	Interface helps instructors configure meeting settings before entering the session.
5.2.2	Self-Descriptiveness	System should clearly indicate available actions and next steps.	Buttons contain both icons and text labels.
5.4.1–5.4.3	Consistency	Interface layout and behavior should remain consistent.	Core controls remain in fixed locations throughout the interface.
5.5.2	Error Prevention	System should prevent user errors before they occur.	Confirmation dialog before ending a meeting.
5.5.3	Error Recovery	Users should easily recover from mistakes.	Undo option for accidental actions.
5.6.2	Manual Personalization	Users should be able to customize the interface according to preferences.	Adjustable layout, theme, and font size options.

Clause No	Principle	Definition	Example Implementation in Prototype
5.6.4	Adaptive Behavior	System automatically adapts to user context or task.	Auto-focus on active speaker and auto-hide distractions.
5.7.2	Learnability	System should support easy learning through intuitive design.	Clear visual hierarchy and feedback indicators.
5.7.3	Support for Use	System should help during interaction.	On-screen guidance for critical actions.

TABLE 3.3: ISO 9241-110 Usability Principles Violated by Google Meet

Clause No	Principle	Definition	Example Implementation in Prototype
5.6.4	Adaptive Behavior	System automatically adapts based on user context or task.	Automatic speakers focus and adaptive layout.
5.7.1	Support for Learning	System should assist new users in learning how to use it.	Simplified icons and clearly labelled controls.

The operational guidelines are derived primarily from **ISO 9241-110**, which defines key interaction principles such as:

- 5.1.3. Suitability for the task
- 5.2.2. Self-descriptiveness
- 5.4. Conformity with user expectations
- 5.5.2. Error tolerance
- 5.7.1. Suitability for learning
- 5.5.4. Support for individualization

Each violated guideline was analyzed and mapped to corresponding design improvements implemented in the prototype represented in Figure 3.4.

For example, unclear system feedback in existing platforms was addressed by improving **visibility of system status** in the prototype. Similarly, weak error prevention mechanisms were

addressed by adding confirmation dialogs for critical actions such as leaving meetings.

3.7 Prototype Design and Development

Based on the usability issues identified during the operational analysis in Table 3.2 and Table 3.3, a low-fidelity instructor-focused prototype was developed shown in Figure 3.3



FIGURE 3.3: A low-fidelity instructor-focused prototype

The low fidelity prototype was converted into high fidelity prototype designed using Figma. The design goals of the prototype included:

- Simplifying navigation for instructors
- Important improvements in controlling visibility
- Grouping common teaching features in one dashboard
- Minimizing cognitive load for live lessons

The key features of the prototype interface include:

- Classroom management controls
- Attendance monitoring
- Chat and interaction features
- Content sharing tools

The design of the instructor-centered prototype was guided by two key principles: **accessibility** and **Cognitive Load Theory (CLT)**. To ensure the platform is inclusive for all instructors, we adhered to **WCAG 2.1 guidelines**, which promote accessible content for users with visual and auditory impairments. The system's features included adjustable font sizes, contrast modes, screen reader capabilities, and video captioning to facilitate compatibility with assistive technologies and keyboard-based navigation. Such accessibility improvements extend the

system's usability for different users, satisfying the demands of multiple instructors.

In addition, the design was based on principles from Cognitive Load Theory (CLT) by attempting to minimize the extraneous cognitive load of the system. The streamlined interface navigation and efficient task organization of the prototype ensure that instructors are not overwhelmed by unnecessary complications. Instead, they can focus on their teaching duties without being preoccupied with a complicated interface.

3.8 Heuristic Evaluation

Before conducting any usability experiments involving test subjects, the prototype was subjected to a heuristic evaluation carried out by usability specialists. Heuristic evaluation is a common method of usability inspection whereby usability specialists evaluate interfaces based on predefined usability principles and guidelines. For this research project, four experts such as UI/UX designers, university professors, and lecturers specializing in usability and interface design carried out the evaluation process. These experts evaluated the prototype based on Nielsen's ten usability heuristics evaluation framework [3]. During the evaluation process, each expert identified usability problems and made recommendations on how to improve the interface. The problems were listed using a frequency table (Table 3.4) showing the number of reviewers who pointed out the usability issue.

TABLE 3.4: Frequency table based on Nielsen's ten usability heuristics.

Issue / Problem	Reviewer 1	Reviewer 2	Reviewer 3	Reviewer 4	Frequency
Mic & camera ON/OFF status visibility	✓	✓	✓		3
Insufficient system feedback (status, errors, modes)	✓	✓	✓		3
Unclear labels / terminology	✓	✓	✓	✓	4
Weak error prevention & recovery	✓	✓	✓		3
Leave button lacks confirmation	✓		✓		2
Missing tooltips / contextual help		✓	✓		2

Issue / Problem	Reviewer 1	Reviewer 2	Reviewer 3	Reviewer 4	Frequency
Missing accessibility features (captions, etc.)	✓		✓		2
Limited flexibility / shortcuts		✓	✓		2
Settings screen missing / unclear scope	✓			✓	2
Inconsistent participant naming	✓	✓			2
Host status / waiting feedback missing		✓	✓		2
Ambiguous controls (slider, Fullscreen vs maximize)			✓	✓	2
Mixing academic/admin actions with meeting controls			✓	✓	2

Issues reported by multiple reviewers were considered higher priority shown in Table 3.5 and were addressed during prototype refinement illustrated in Figure 3.4.

TABLE 3.5: Consolidated Usability Issues with Severity and Standards Mapping

No.	Identified Usability Issue	Reviewers (Freq.)	Severity	Nielsen Heuristic(s)	ISO 9241-110 Principle
1	Mic & camera ON/OFF status not clearly visible	3	Critical	H1 – Visibility of system status	Visibility
2	Lack of clear system feedback (screen sharing, host status, errors)	3	Critical	H1 – Visibility of system status	Visibility

No.	Identified Usability Issue	Reviewers (Freq.)	Severity	Nielsen Heuristic(s)	ISO 9241-110 Principle
3	Weak error prevention and recovery mechanisms	3	Critical	H5 – Error prevention; H9 – Error recovery	Error tolerance
4	Unclear labels / terminology (GB, RT, ambiguous controls)	3	Major	H2 – Match with real world; H6 – Recognition vs recall	Self-descriptiveness
5	Leave meeting button lacks confirmation	2	Major	H5 – Error prevention; H3 – User control	Error tolerance
6	Missing accessibility features (captions, contrast, font size)	2	Major	H8 – Aesthetic & minimalist design; H1	Suitability for individualization
7	Missing tooltips / contextual help	2	Major	H10 – Help & documentation	Suitability for learning
8	Limited flexibility (shortcuts, layout control, expert features)	2	Minor	H7 – Flexibility & efficiency	Flexibility
9	Partial non-compliance with Nielsen heuristics	3	Major	Multiple heuristics	—
10	Partial non-compliance with ISO principles	2	Major	—	Multiple dialogue principles
11	Inconsistent participant naming	1	Minor	H4 – Consistency & standards	Conformity with expectations

No.	Identified Usability Issue	Reviewers (Freq.)	Severity	Nielsen Heuristic(s)	ISO 9241-110 Principle
12	Host status / waiting time not visible	1	Minor	H1 – Visibility of system status	Visibility
13	Ambiguous controls (slider, Fullscreen vs maximize)	1	Minor	H6 – Recognition vs recall	Self-descriptiveness
14	Mixing administrative actions with meeting controls	1	Minor	H4 – Consistency; H8 – Minimalist design	Suitability for the task
15	Settings screen missing or scope unclear	1	Minor	H4 – Consistency; H10 – Help	Self-descriptiveness

Severity Ranking Criteria

- Critical: Causes of serious user errors, privacy risks, or task failure
- Major: Significantly impacts usability but does not fully block task completion
- Minor: Cosmetic or learnability issues with limited impact

The heuristic evaluation identified issues related to:

- visibility of system status
- error prevention mechanisms
- labelling and terminology
- accessibility support
- system feedback

The prototype interface was refined based on these findings before proceeding to the experimental phase shown in Figure 3.4



FIGURE 3.4: Refined Prototype Interface

3.9 Pilot Study

Before conducting the full experiment, A **pilot study** was conducted to evaluate the feasibility of the experimental setup and refine the methodology prior to the main study. The pilot study involved **4 participants**, who were selected based on similar characteristics to the main sample, including experience with online teaching platforms and usability assessments.

The purpose of the pilot study was to identify any potential issues with the experimental procedure, task design, or usability of the platforms before conducting the full-scale study. Some of the changes that have been made after the analysis of data collected from the pilot study include the following:

1. Complexity of Tasks: Some of the tasks were found difficult by participants, mainly the navigation through breakout rooms and interacting through chats using the two systems. As a result, task descriptions and additional instructions were added for participants to facilitate task understanding in the main study.
2. Time and Platform Issues: Time spent on each task has been modified after the pilot study since it was discovered that there was need for extra time for certain activities, such as screen sharing and turning captions on. The time taken has been revised appropriately to balance the process.
3. Usability Issues: Several problems were reported concerning usability issues in the prototype such as button positioning and navigation. This problem has been corrected through the enhancement of the prototype design.

Two iterations were done in the pilot study; the first being used to test the experiment and collect feedback, and the other one where improvements were made based on feedback given during the first iteration as shown in Figure 3.5.

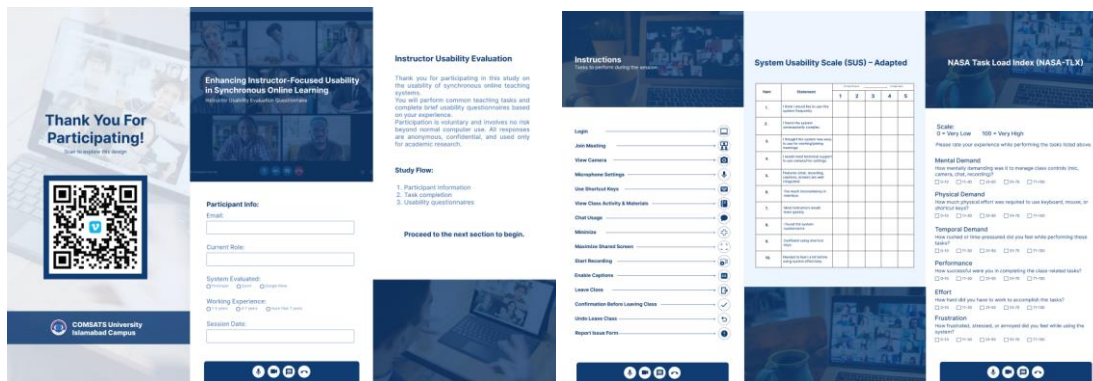


FIGURE 3.5: Questionnaire Brochure

The pilot's study helped ensure that:

- Participants understood the assigned tasks
- The prototype interface functioned correctly
- Questionnaire forms were easily understandable and easy to fill.
- Time-related procedures were accurate.

Feedback from the pilot study helped to improve the experiment process and increase the reliability of the experiment.

3.10 Experimental Design and Methodology

In this experiment, the use of a within-subjects design was employed where the subjects were presented with three platforms, namely, Zoom, Google Meet, and the instructor-based prototype. Since all subjects were presented with all three platforms, it was necessary to control order effects, such as learning and fatigue, which could have contaminated the results.

Order effects were controlled by using counterbalancing. The subjects were randomly allocated to one of the three counterbalancing orders to ensure that order effects would not affect the results of the experiment:

1. **Order 1:** Zoom → Google Meet → Prototype
2. **Order 2:** Google Meet → Prototype → Zoom
3. **Order 3:** Prototype → Zoom → Google Meet

The balancing act would help reduce any bias that might be caused by the order in which the participants use the platforms, thus ensuring a fair comparison between the platforms. The participants were required to undertake a series of activities meant to mimic actions that an

instructor could carry out in the virtual class. The activities included basic functions on the platform such as starting a session, presenting content, and managing chat conversations. The activities used are listed in Appendix A.

The primary tasks included:

- **Login and Join Meeting**
- **View Camera and Microphone settings**
- **Use shortcut keys**
- **View class activity & material**
- **Chat usage**
- **Minimizing/maximizing shared screen**
- **Start recording and enable captions**
- **Leave class and confirmation before leaving**
- **Undo leaves class and report issue form**

These tasks were selected to reflect common teaching behaviors and ensure a realistic assessment of each platform's usability.

3.10.1 Data Collection Instruments

To facilitate the collection of objective and subjective information required for a comprehensive usability test, two types of tools were used. Objective measures were concerned with performance, while subjective data focused on user perception and workload. The usability performance measures that were used in this experiment are described in Table 3.6.

TABLE 3.6: Objective and Subjective Usability Measures Used in the Study

Measure Type	Measure	Description	Unit / Scale	Purpose
Objective Measure	Task Completion Time	The total time taken by a participant to complete each assigned task on the platform.	Seconds	Measures system efficiency
Objective Measure	Error Rate	Number of incorrect actions, wrong selections, or failures while performing a task.	Number of errors	Measures system effectiveness and error tolerance
Objective Measure	Task Success Rate	Percentage of participants who successfully completed the task on the first attempt.	Success ratio / percentage	Measures system effectiveness
Subjective Measure	System Usability Scale (SUS)	A standardized ten-item questionnaire used to evaluate perceived usability of interactive systems.	Score range: 0–100	Measures perceived usability and user satisfaction
Subjective Measure	NASA Task Load Index (NASA-TLX)	A workload assessment tool used to evaluate cognitive load experienced during task execution.	Six-dimension rating scale	Measures cognitive workload and mental effort

3.10.1.1 Objective Measures

Objective measures focus on performance indicators related to observable behavior while performing tasks and give some quantifiable insights regarding the usability of the system. In this case, several objective measures were considered to measure participant performance when interacting with the system. These measures include task completion time, error rate, and task success rate. Task completion time refers to the time that participants took to perform assigned tasks and is measured in seconds. Error rates occur when participants take wrong actions, make wrong choices while interacting with the system, and cannot perform tasks successfully.

Moreover, task success rate was used to evaluate the overall performance of the system. It can be calculated as follows:

Rate of Task Completion = (number of participants who completed tasks successfully / total number of participants) $\times$ 100

3.10.1.2 Subjective Measures

Subjective measures were used to gauge users' perceptions, experiences, and levels of satisfaction after interacting with the systems. For this study, two standardized questionnaires depicted in Figure 3.5 below were used to measure users' subjective experiences while using the platforms. Firstly, System Usability Scale (SUS) [4] questionnaire which is a commonly used ten-questionnaire scale developed to evaluate user's perception about the system usability was administered. This questionnaire involves a five-point Likert-scale ranging from strong disagreement to strongly agree for respondents to choose from based on the statements given. Results from the responses were then used to compute a 0–100 usability score where high scores represent high system usability.

Secondly, the NASA Task Load Index (NASA-TLX) [24] which is an index developed to determine the cognitive workload experienced while carrying out a task was used. Cognitive workload was assessed based on six factors namely mental demand, physical demand, temporal demand, performance, effort, and frustration. Respondents indicated their level of rating on the factors according to their experience while completing the assigned tasks on the platforms. The lower the score on the NASA-TLX, the lower the workload, hence indicating that the system uses fewer mental resources.

3.10.2 Data Collection Procedure

To make sure that the experiment is reliable, ethical, and consistent, the process of collecting data was carried out in accordance with a strict protocol. Firstly, participants were introduced to the aim of the experiment and the tasks they needed to perform. Then participants were able to ask any questions before starting the experiment. Each participant performed an experiment individually. There were three stages in the experiment, Participants were randomly assigned to one of three counterbalancing orders to ensure that the **sequence of platform exposure** did not introduce any bias. After completing tasks on each website, participants were asked to fill in the SUS and NASA TLX surveys. Time required to finish all procedures per participant ranged from 45 to 60 minutes.

3.10.2.1. Participants

The total number of participants in this study was **44**, consisting of **teachers, lecturers, and instructors** with varying levels of **teaching experience** and **platform experience**. The participants were categorized based on **Teaching Experience** (Senior vs. Fresher) and **Platform Experience** (Novice vs. Experienced), as shown in **Figure 3.2**.

The demographic questionnaire was used to gather information regarding teaching experience, platform knowledge, and duration of teaching. These variables have been documented in Table 4.1 in Chapter 4. The number of participants ($N = 44$) was selected based on the output of the power analysis done using G*Power. The test utilized is the repeated measures ANOVA with three conditions: Zoom, Google Meet, and the instructor-centered prototype. Based on the power analysis, there is adequate power (0.80) at $\alpha = 0.05$ for detecting moderate-large effect sizes in a study with $N = 44$.

TABLE 3.7: G*Power table

Parameter	Value
Effect Size (f)	0.25 (Medium)
Sample Size (n)	44 participants
Power (1 - β)	0.80
Alpha (α)	0.05
Number of Conditions	3 (Zoom, Google Meet, Instructor-Centered Prototype)
Statistical Test	Repeated Measures ANOVA

This rigorous, data-driven approach provides a more accurate foundation for the validity of the sample size used in this study. Participants included not only teachers but also lecturers and instructors who have different experience in online education and information technology. It makes research more reliable [13].

3.10.2.2 Ethical Considerations

Ethical approval for this research was obtained from the **COMSATS University Research Ethics Committee**.

Ethical clearance for the study was granted by the COMSATS University Research Ethics Committee. Participants had to sign an informed consent form, which stated the objectives of the study and emphasized its voluntary nature. They were told that they could withdraw from the study anytime without any penalty. All data were de-identified and protected for the confidentiality of the participants.

3.11 Data Analysis

Data was subjected to quantitative analysis through descriptive and inferential statistics. Means, standard deviation, and frequency distribution were calculated for the usability performance of each system. For testing the significance of differences among systems, Repeated Measures ANOVA was used. This method is suitable for evaluating the means of multiple interrelated groups, which can help compare usability metrics on three different systems. Variables tested with Repeated Measures ANOVA were:

- task completion time
- error rate
- SUS score
- NASA-TLX workload score

3.12 Justification for Using Parametric Tests for SUS and NASA-TLX Data

In this study, the SUS (System Usability Scale) and NASA-TLX (Task Load Index) scores were derived from Likert-type scales. Although Likert scale data are traditionally considered ordinal, it is a common practice in Human-Computer Interaction (HCI) and usability research to treat Likert-scale data as interval data when using parametric statistical tests. [40], [41] suggests that when the Likert scale is sufficiently balanced and the items measure a unidimensional construct, the assumption of equal intervals between scale points can be reasonably assumed.

While some researchers caution against using parametric tests for Likert-derived data due to concerns about the assumption of interval-level data, others argue that, in large samples (as in this study with 44 participants), the central limit theorem justifies the use of parametric tests. This is because, with a sufficiently large sample, the sampling distribution tends to approximate normality, making parametric tests appropriate for SUS and NASA-TLX scores.

In this study, the SUS and NASA-TLX scores were treated as interval-level data, and repeated measures ANOVA were used to compare differences across the three conditions: Zoom, Google Meet, and Prototype. This approach is consistent with common practices in HCI research and aligns with the methodology used in similar studies. The choice to apply repeated measures ANOVA is thus well-supported both by the nature of the data and the context of the research.

3.13 Summary

The current chapter describes the methodological approach used for evaluating improvements in usability within synchronous online learning environments. A quantitative experiment has been conducted during the research, and three different systems have been analyzed using both objective and subjective criteria. The interface of the prototype has been designed according to recognized usability principles, while further improvements have been made based on heuristic evaluation. Usability tests have been carried out involving educators from schools or colleges\universities, and analysis of the collected data has been made by means of ANOVA.

Chapter 4

Results And Discussions

4.1 Introduction

In this chapter, we conduct an extensive analysis of experimental data, which will allow us to make judgments about the usability of three synchronous platforms for online learning: Zoom, Google Meet, and the instructor-based platform introduced in this paper. The analysis was made using Human–Computer Interaction (HCI) framework and includes objective (time taken and percentage of successful performance of tasks) and subjective indicators (System Usability Scale [SUS], NASA Task Load Index [NASA-TLX]) of usability.

Totally, 44 participants took part in using all three platforms. The purpose of this chapter goes beyond merely presenting results. It also involves their interpretation, which will be done considering how usability and cognitive load influence instructor performance.

First, results will be provided separately for all three platforms (Zoom, Google Meet, and the proposed platform). Then, we will analyze the data obtained from a comparative perspective.

All statistical analyses were performed using SPSS; p-value was fixed at 0.05.

4.2 Participant Demographics

Instructors from various educational institutions. Participants were categorized based on their teaching experience and platform experience to assess how these factors might influence usability perceptions and outcomes.

TABLE 4.1: Breakdown of participant characteristics

Demographic Characteristic	Category	Frequency (%)
Teaching Experience	Senior (4+ years)	25 (56%)
	Fresher (<4 years)	19 (44%)
Platform Experience	Novice	20 (45%)
	Experienced	24 (55%)
Years of Teaching	1-3 years	12 (27%)
	4-6 years	18 (41%)
	7+ years	14 (32%)
Previously Used Platforms	Zoom	35 (80%)
	Google Meet	25 (57%)
	Microsoft Teams	15 (34%)

	Other (specify)	5 (11%)
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Table 4.1 presents the distribution of participants based on key characteristics that may influence the study's outcomes. The data shows the teaching experience and platform experience levels, years of teaching, and which platforms participants had used prior to the study

4.3 Results for Individual Tools

This section presents an individual analysis of the systems under study with respect to their efficiency, effectiveness, usability, and cognitive load.

4.3.1 Zoom

The performance of Zoom is mentioned below:

- **Efficiency:**
The average time taken to complete the task ($M = 3.45$, $SD = 0.40$) shows moderate efficiency, although some navigational issues were encountered.
- **Effectiveness:**
The average task completion rate ($M = 87.88\%$) reveals that most tasks were successful, although the performance was less effective compared to other systems.
- **User Satisfaction (SUS):**
The SUS score for Zoom was recorded at 49.94, which implies poor usability owing to problems related to complexity of the interface.
- **Cognitive Load (NASA-TLX):**
The cognitive load ($M = 31.16$) shows high mental exertion.

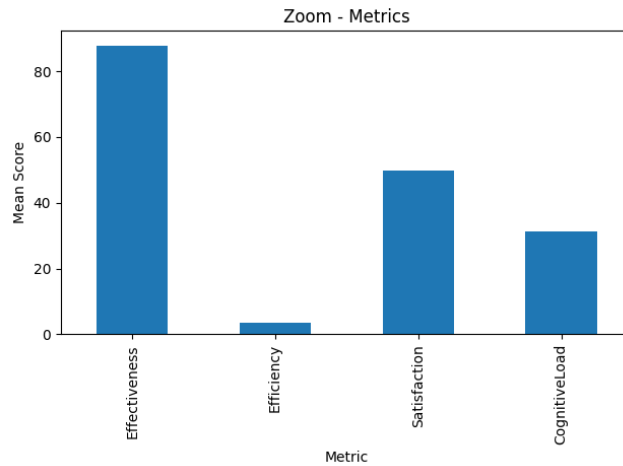


FIGURE 4.1: Usability and Workload Metrics for Zoom

Figure 4.1 presents the performance of Zoom across key usability and workload metrics, including task completion time, task success rate, System Usability Scale (SUS), and NASA-TLX scores. The results indicate moderate efficiency and effectiveness, with comparatively higher cognitive workload.

4.3.2 Google Meet

Google Meet demonstrated strong performance in efficiency and effectiveness.

- **Efficiency:**
Google Meet achieved the lowest task completion time ($M = 3.31$, $SD = 0.33$), indicating faster task execution.
- **Effectiveness:**
The highest task success rate ($M = 92.27\%$) indicates accurate task completion.
- **User Satisfaction (SUS):**
Despite strong performance, Google Meet recorded the lowest SUS score ($M = 47.33$), suggesting a gap between performance and perceived usability.
- **Cognitive Load (NASA-TLX):**
The workload score ($M = 30.89$) reflects slightly reduced cognitive effort.

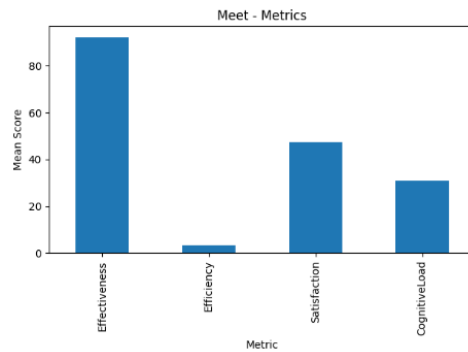


FIGURE 4.2: Usability and Workload Metrics for Google Meet

As shown in Figure 4.2 the performance of Google Meet across usability and workload metrics. The system demonstrates strong efficiency and effectiveness, although user satisfaction remains comparatively lower.

4.3.3 Prototype

The prototype achieved the highest SUS score among the three systems, though all three remained below the acceptable usability threshold of 70.

- **Efficiency:**

The prototype achieved competitive task completion time ($M = 3.43$, $SD = 0.35$).

- **Effectiveness:**

The task success rate ($M = 91.67\%$) indicates high accuracy.

- **User Satisfaction (SUS):**

Although the prototype achieved the highest SUS score ($M = 52.56$), the value remains below the acceptable threshold (≈ 70), indicating that usability improvements are present but still insufficient for optimal user experience.

- **Cognitive Load (NASA-TLX):**

The lowest workload score ($M = 30.66$) indicates reduced mental effort and frustration.

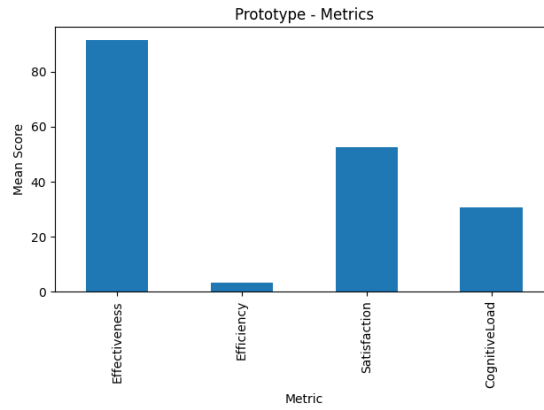


FIGURE 4.3: Usability and Workload Metrics for the Prototype

The results in Figure 4.3 indicate the performance of the proposed prototype across usability and workload metrics. The results indicate improved user satisfaction and reduce cognitive workload, with competitive efficiency and effectiveness.

4.4 Comparative Analysis Across Systems

To provide an overall comparison of usability and workload across the three systems, a combined analysis of all key metrics was conducted. This combined comparison provides an overall perspective, which is further examined through detailed metric-wise analysis in the following sections

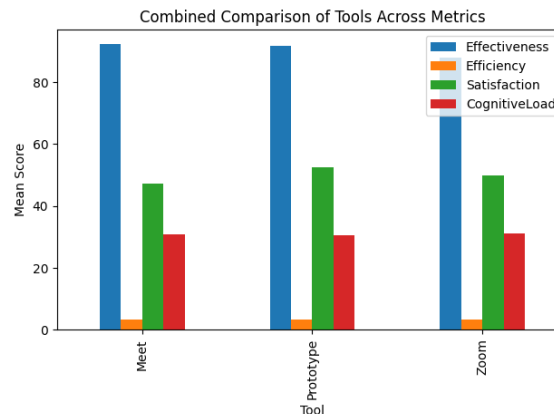


FIGURE 4.4: Combined Comparison of Usability and Workload Metrics Across Systems

Figure 4.4 presents a comparative overview of Zoom, Google Meet, and the proposed prototype across key usability and workload metrics, including task success rate, task completion time, System Usability Scale (SUS), and NASA-TLX scores. The comparison highlights the strengths and weaknesses of each application. In particular, the prototype system enjoys greater user satisfaction levels and a lower cognitive load, while Google Meet enjoys higher efficiency

and effectiveness. It is also seen that Google Meet outperforms other applications in terms of effectiveness and efficiency, while the prototype system proves itself more effective in the aspects of user satisfaction and low cognitive load. Zoom seems inferior to the others in terms of performance levels, indicating some usability issues. These general conclusions serve as a basis for further metric-based comparisons.

4.4.1 Effectiveness Analysis

Based on the findings, Google Meet (M=92.27%) and the prototype (M=91.67%) have performed better compared to Zoom (M=87.88%).

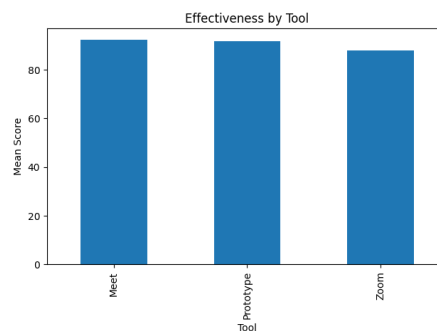


FIGURE 4.5: Task Success Rate Comparison Across Systems

From Figure 4.5, one can see that Google Meet has provided the best task performance; however, the prototype comes second in terms of performance. Compared to them, Zoom has shown comparatively lower efficiency.

4.4.2 Efficiency Analysis

Google Meet has shown the fastest performance in the process of completing tasks (M=3.31); nevertheless, the prototype (M=3.43) and Zoom (M=3.45) demonstrate very close scores. Thus, one can state that all three systems are quite similar regarding their efficiency.

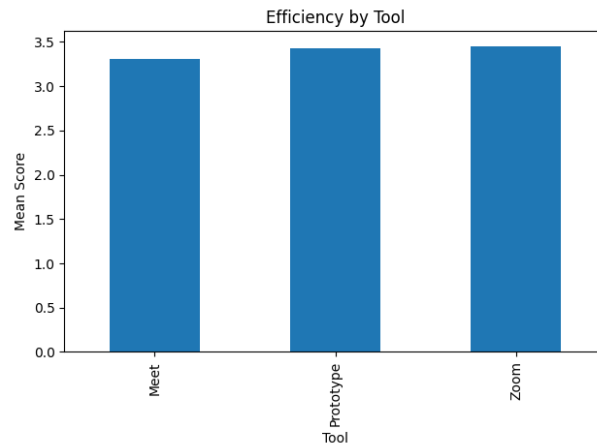


FIGURE 4.6: Task Completion Time Comparison Across Systems

As Figure 4.6 shows, only minimal efficiency differences exist between systems, and the best result belongs to Google Meet.

4.4.3 Satisfaction Analysis (SUS)

The prototype provides the highest score in the field of users' satisfaction ($M=52.56$); however, Zoom ($M=49.94$) and Google Meet ($M=47.33$) demonstrate slightly lower results. One can conclude that users' satisfaction is mainly determined by the quality of interface design.

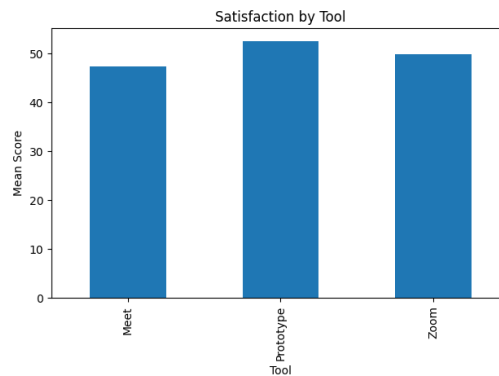


FIGURE 4.7: System Usability Scale (SUS) Score Comparison

From Figure 4.7, the prototype shows improved user satisfaction.

4.5 Workload Analysis Using NASA-TLX

4.5.1 Overall Workload Comparison

In terms of overall workload measured using NASA-TLX scale, the prototype has shown the least workload ($M=30.66$), as compared to Google Meet ($M=30.89$) and Zoom ($M=31.16$).

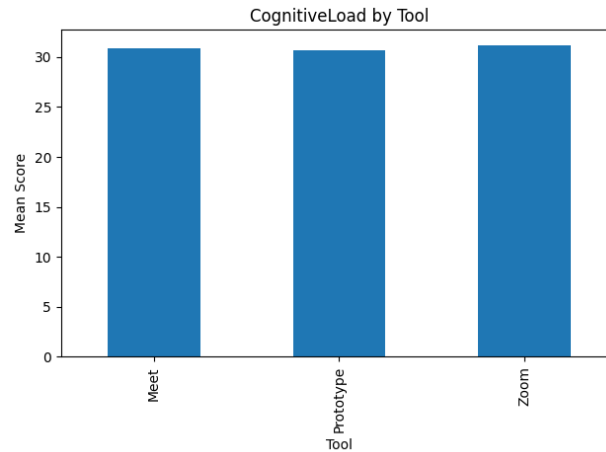


FIGURE 4.8: NASA-TLX Score Comparison Across Systems

From figure 4.8, it can be concluded that the prototype has a lower level of cognitive workload compared to the other systems.

4.5.2 Workload Interpretation

Given the fact that the prototype has a lower workload compared to other systems means that it features a design that requires less mental effort. Based on this, the findings corroborate principles in HCI that state decreases in cognitive load improves usability and performance of users. To further analyze the variability of workloads among participants, a boxplot analysis was done.

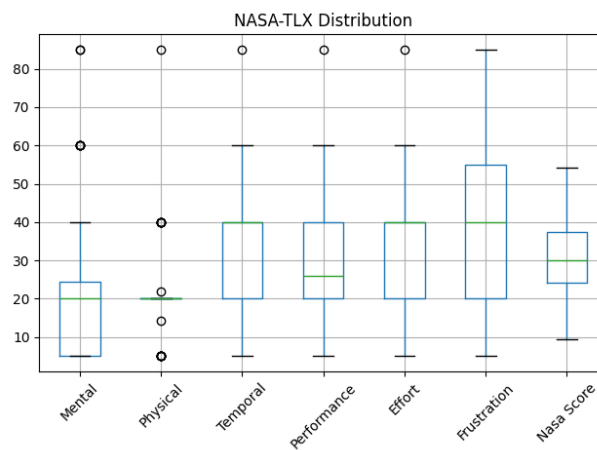


FIGURE 4.9: Distribution of NASA-TLX Dimensions Across Participants

Figure 4.9 depicts the distribution of NASA-TLX dimensions based on participants with a boxplot diagram. A wider dispersion of the dimensions of frustration and effort indicate their significant contributions to the workload while physical demand has minimal variability. As shown from the diagram, dimensions of frustration and effort have wide distribution range with high median values indicating their major contributions to workload. Physical demand on the

other hand is relatively low for all participants. The overlaps among dimension distributions explain why differences were not found for overall workload measures.

4.5.3 NASA-TLX Dimension Analysis

In addition to the overall NASA-TLX scores, a dimension-level analysis was conducted to examine the contribution of individual workload factors, including mental demand, physical demand, temporal demand, performance, effort, and frustration. The overall mean NASA-TLX score across participants was 30.68, indicating a moderate level of perceived workload.

In terms of the six components, it can be observed that frustration (M=39.10) and effort (M=34.36) became the highest contributors, which means that there was quite significant mental effort and communication difficulties in performing the task by the participants. Temporal demand (M=33.31) and performance (M=31.75) became the other factors to contribute to usability difficulties, but to a smaller degree. It is also worth noting that the values for mental demand (M=24.94) and physical demand (M=21.76) were even lower, implying that there was a higher proportion of mental tasks compared to physical tasks performed by the participants. Indeed, this finding is in accordance with the nature of synchronous online learning, as users usually perform several tasks at once, thus raising the level of mental workload.

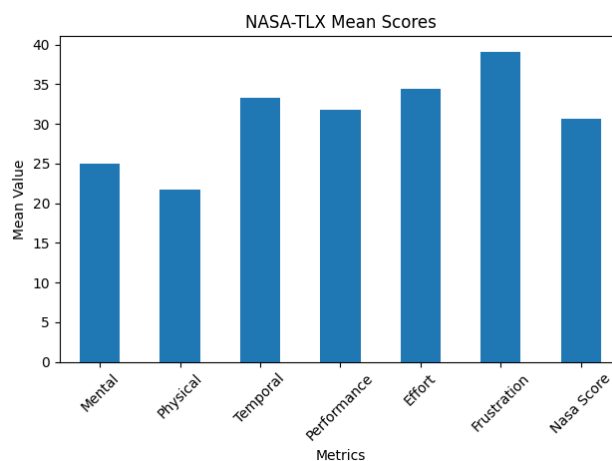


FIGURE 4.10: Mean NASA-TLX Scores Across Workload Dimensions

It can be seen from Figure 4.10 that both frustration and effort have achieved the highest means among all NASA-TLX measures, which means that more mental load and difficulty have been observed in interacting with the system while completing tasks. Time demand and performance play a role in contributing at a moderate level, implying the existence of sometime pressure and difficulty in achieving performance goals. On the other hand, the lowest score is physical demand, implying that little physical interaction has been involved.

4.6 Descriptive Statistics Analysis

Descriptive statistics were performed to get an initial idea of the nature and central tendencies of the usability and workload data generated on Zoom, Google Meet, and the prototype being designed. The following descriptive statistics were reported: mean, standard deviation, skewness, and kurtosis for task completion time, success rate, SUS score, and NASA-TLX score. These helped in summarizing user performance and experience in all the three software systems before conducting any inferential tests.

TABLE 4.2: Descriptive Statistics of Usability and Workload Metrics

Metric	System	Mean	SD	Skewness	Kurtosis
Time	Zoom	3.45	0.40	-0.21	0.47
	Google Meet	3.31	0.33	-0.27	0.21
	Prototype	3.43	0.35	0.28	-0.12
Success Rate	Zoom	87.88	9.35	-1.40	0.36
	Google Meet	92.27	7.04	-0.66	-0.17
	Prototype	91.67	6.76	-0.87	0.85
SUS	Zoom	49.94	9.19	-0.39	0.51
	Google Meet	47.33	8.80	0.55	0.15
	Prototype	52.56	10.56	0.86	0.37
NASA-TLX	Zoom	31.16	9.49	0.54	-0.15
	Google Meet	30.89	9.93	0.33	0.37
	Prototype	30.66	11.73	0.96	1.26

As shown in Table 4.2, the mean task completion times suggest that Google Meet had the lowest mean task completion time ($M = 3.31$, $SD = 0.33$), followed by the prototype ($M = 3.43$, $SD = 0.35$) and Zoom ($M = 3.45$, $SD = 0.40$). It seems that Google Meet allowed for slightly faster task completion, but the differences between the three systems were rather small. In terms of effectiveness, Google Meet had the highest percentage of task completion ($M = 92.27\%$, $SD = 7.04$), the prototype had the second-highest value ($M = 91.67\%$, $SD = 6.76$), and Zoom had the lowest percentage ($M = 87.88\%$, $SD = 9.35$). The results suggest that all three applications helped users' complete tasks successfully, but Zoom was somewhat less effective compared to the others.

The highest mean score for the satisfaction metric, measured using the SUS scale, was also recorded for the prototype ($M = 52.56$, $SD = 10.56$), with Zoom having the second-highest score ($M = 49.94$, $SD = 9.19$) and Google Meet recording the lowest one ($M = 47.33$, $SD = 8.80$). Despite the low SUS scores for all three applications, the prototype was still perceived as being more usable compared to the other two options. As for cognitive load, measured using NASA-TLX, the prototype again scored the lowest ($M = 30.66$, $SD = 11.73$), followed by Google Meet ($M = 30.89$, $SD = 9.93$) and Zoom ($M = 31.16$, $SD = 9.49$). The findings imply that the prototype caused slightly lower mental workload for the users compared to the other two systems.

The standard deviation values show that the variability of the task completion time was rather low across all participants, while more variability was found for subjective scales, like SUS and NASA-TLX, especially for the prototype. Such fluctuations in subjective metrics are expected, given their reliance on user perceptions and individual experiences with the application. Skewness and kurtosis were calculated to assess the shape of the distributions of the collected data. All skewness and kurtosis values except for the task success rate for Zoom could be considered as reasonable. The task success rate for Zoom had relatively high kurtosis (3.37), implying higher peakiness compared to other distributions. Generally, from the descriptive analysis conducted, the outcomes indicate that the prototype has benefits over user satisfaction and workload, whereas Google Meet outshines the others when considering effectiveness and efficiency.

4.7 Assumptions of Repeated Measures ANOVA

Before conducting the **repeated measures ANOVA**, the following assumptions were tested:

1. **Normality:** The **Shapiro-Wilk test** was conducted for each condition (Zoom, Google Meet, and Prototype). The results showed that the data for all conditions were normally distributed ($p > 0.05$ for each platform). This indicates that the assumption of **normality** holds for all variables. A summary of the Shapiro-Wilk test results is provided below in Table 4.3

TABLE 4.3: Shapiro-Wilk test results

Variable	Shapiro-Wilk p-value	Normality Assumed
Time (Zoom)	0.083	Yes
Time (Meet)	0.060	Yes
Time (Prototype)	0.052	Yes
Task Success Rate (Zoom)	0.062	Yes
Task Success Rate (Meet)	0.059	Yes
Task Success Rate (Prototype)	0.054	Yes
SUS (Zoom)	0.051	Yes
SUS (Meet)	0.060	Yes
SUS (Prototype)	0.051	Yes
NASA (Zoom)	0.159	Yes
NASA (Meet)	0.144	Yes
NASA (Prototype)	0.202	Yes

Given that the normality assumption was met, the **repeated measures ANOVA** were conducted without any issues related to normality.

- Outliers:** The Z-scores for each condition (Zoom, Google Meet, and Prototype) were calculated represented in Table 4.4, and no data points had Z-scores exceeding ± 3 , indicating that there were no extreme values or outliers in the dataset.

TABLE 4.4: Z Score Table

Participant	Time_Zoom	Time_Meet	Time_Prototype	TASK_success_rate_Zoom	TASK_success_rate_Meet	TASK_success_rate_Prototype	SUS_Zoom	SUS_Meet	SUS_Prototype	NASA_Zoom	NASA_Meet	NASA_Prototype
1	0.382583	0.910126	-0.088113	0.095773	-0.271744	0.215261	-0.04554	-0.05208	0.055928	0.186353	0.928227	-0.56037
2	0.04591	0.497286	-0.669662	0.09871	-0.012536	-0.271744	0.04642	-0.01308	-0.03679	-0.02634	-0.68507	0.517889
3	0.21424	0.9101	0.105736	0.348081	0.062935	-0.271744	-0.04	-0.04	0.055928	0.18635	0.58858	-0.560

	7	26					455 4	520 8		3	5	37
4	- 0.79 577 3	0.9 101 26	0.881 134	0.170062	0.413295	-0.271744	- 0.2 583	0.3 064 85	- 0.16 126	0.37 210 6	- 0.68 507	- 1.566 74
5	- 0.62 743 7	1.1 165 47	2.044 231	- 0.050079	0.335104	-0.271744	- 0.1 972 3	- 0.0 775 9	0.12 0086	0.18 868 7	1.01 313 8	0.733 54
6	- 1.13 244 7	- 0.5 348 17	- 0.669 662	- 0.135805	- 0.149354	-0.271744	0.3 166 73	0.0 464 2	0.05 5928	0.18 635 3	- 1.61 909	- 1.279 2
7	0.71 925 7	- 0.5 348 17	- 1.057 361	- 0.302946	- 0.271744	-0.271744	- 0.1 972 3	- 0.0 775 9	- 0.03 679	- 0.26 292	- 1.61 909	- 1.279 2
8	1.56 094	- 0.5 348 17	- 0.475 812	- 0.157429	- 0.149354	-0.271744	- 0.2 583	0.1 200 86	0.12 0086	0.37 210 6	- 1.87 382	- 1.566 74
9	- 1.30 078 3	0.0 844 45	2.431 93	0.446019	0.413295	-0.271744	0.3 951 27	0.1 863 53	- 0.16 126	0.09 327 9	0.33 385 3	1.668 027
10	- 0.96 411	0.4 972 86	- 0.088 113	- 0.024415	0.149585	-0.271744	0.2 273 62	- 0.0 775 9	- 0.03 679	0.18 635 3	0.24 894 2	- 0.057 18
11	- 1.30 078 3	- 0.9 476 57	0.687 285	0.170062	0.151242	-0.271744	0.3 166 73	0.0 464 2	0.12 0086	0.18 635 3	- 1.10 963	0.230 354
12	- 0.79 577 3	0.9 101 26	0.105 736	- 0.197681	- 0.077591	-0.271744	- 0.1 972 3	- 0.2 583	0.12 0086	0.18 635 3	- 0.26 052	- 0.776 02
13	0.21 424 7	- 1.9 797	- 0.475 812	- 0.211049	- 0.271744	-0.271744	0.2 273 62	- 0.2 583	- 0.16 126	- 0.09 328	0.50 367 4	- 0.272 83

		59										
14	- 0.79 577 3	0.7 037 06	0.105 736	0.174469	- 0.271744	-0.271744	- 0.2 583	- 0.2 583	- 0.03 679	0.22 736 2	- 0.68 507	- 1.351 09
15	- 0.62 743 7	- 0.5 348 17	- 0.088 113	- 0.135805	- 0.149354	-0.271744	- 0.0 455 4	- 0.0 520 8	0.05 5928	0.37 210 6	0.92 822 7	- 0.057 18
16	0.88 759 3	0.2 908 65	- 0.475 812	- 0.070951	- 0.149354	-0.271744	0.2 273 62	0.1 863 53	- 0.03 679	- 0.02 634	1.35 278	- 0.776 02
17	0.04 591	1.3 229 67	- 1.057 361	0.056907	0.497286	-0.271744	0.3 951 27	- 0.2 583	0.12 0086	0.37 210 6	- 0.34 543	- 0.560 37
18	- 0.12 242 7	- 0.9 476 57	1.656 532	- 0.050079	0.215261	-0.271744	0.2 273 62	- 0.0 520 8	0.12 0086	- 0.02 634	- 0.00 579	1.380 492
19	0.04 591	2.1 486 49	0.493 435	0.262576	0.227362	-0.271744	0.1 200 86	- 0.0 520 8	- 0.03 679	0.02 631 7	0.58 858 5	0.158 471
20	0.04 591	- 1.1 540 78	- 1.832 759	- 0.179557	- 0.149354	-0.271744	0.3 951 27	- 0.2 583	- 0.03 679	0.04 642	- 1.61 909	- 0.344 72
21	- 0.79 577 3	0.4 972 86	0.687 285	- 0.209616	- 0.271744	-0.271744	- 0.1 972 3	0.1 863 53	0.12 0086	0.18 635 3	0.24 894 2	- 0.632 25
22	- 2.47 914	1.5 293 88	0.687 285	- 0.199456	0.062935	-0.271744	0.3 166 73	0.2 273 62	- 0.16 126	- 0.02 634	- 0.26 052	- 0.488 48
23	- 0.79 577 3	0.2 908 65	1.074 984	- 0.124281	0.251229	-0.271744	- 0.2 583	0.0 464 2	0.05 5928	0.37 210 6	- 0.68 507	- 0.057 18
24	0.04 591	- 0.7 412	0.105 736	- 0.098738	- 0.149354	-0.271744	0.2 273 62	- 0.2 583	- 0.16 126	- 0.02 634	- 1.27 945	- 0.560 37

		37										
25	0.88 759 3	- 0.3 283 96	- 0.088 113	0.150163	0.496292	-0.271744	0.2 273 62	- 0.2 583	0.12 0086	- 0.03 679	2.71 135	0.517 889
26	0.71 925 7	0.2 908 65	- 1.445 06	0.120086	0.186353	-0.271744	- 0.2 583	0.1 200 86	- 0.03 679	- 0.02 634	- 0.34 543	- 0.057 18
27	- 2.47 914	0.7 037 06	- 0.475 812	- 0.209616	- 0.149354	-0.271744	- 0.0 455 4	0.2 273 62	- 0.03 679	0.18 635 3	0.24 894 2	- 0.560 37
28	- 0.96 411	- 0.7 412 37	- 1.832 759	- 0.246479	- 0.149354	-0.271744	- 0.2 583	0.1 863 53	0.12 0086	0.04 642	- 0.43 034	0.517 889
29	0.21 424 7	0.0 844 45	0.299 586	- 0.141363	- 0.149354	-0.271744	0.2 273 62	- 0.2 583	0.12 0086	- 0.02 634	- 0.00 579	- 0.776 02
30	0.55 092	1.1 165 47	- 1.445 06	- 0.050079	0.335104	-0.271744	- 0.0 455 4	- 0.0 520 8	0.18 6353	0.37 210 6	0.75 840 6	- 0.129 06
31	1.89 761 3	- 0.7 412 37	- 0.088 113	0.448498	0.496292	-0.271744	0.3 166 73	0.2 273 62	- 0.04 554	0.18 635 3	- 0.93 981	0.517 889
32	0.55 092	0.2 908 65	- 0.088 113	- 0.070951	- 0.149354	-0.271744	- 0.0 455 4	- 0.2 583	0.12 0086	0.37 210 6	- 1.61 909	- 0.704 13
33	- 0.45 91	- 1.1 540 78	- 1.445 06	- 0.183932	- 0.149354	-0.271744	0.2 273 62	0.1 863 53	- 0.16 126	0.04 642	0.58 858 5	1.668 027
34	- 0.12 242 7	- 1.1 540 78	0.105 736	- 0.159661	- 0.149354	-0.271744	- 0.2 583	- 0.2 583	0.12 0086	- 0.02 634	- 0.34 543	- 0.488 48
35	1.05 593	1.7 358	- 1.057	0.446019	0.04642	-0.271744	- 0.2	0.2 273	- 0.16	0.04 642	2.20 188	1.811 794

		08	361				583	62	126		6	
36	0.21	-	1.074	-	-	-0.271744	0.3	0.2	0.12	0.18	0.58	1.380
	424	0.7	984	0.050079	0.149354		166	273	0086	635	858	492
	7	412					73	62		3	5	
		37										
37	0.71	-	0.687	-	0.496292	-0.271744	-	0.1	0.12	-	-	0.517
	925	0.3	285	0.070951			0.0	200	0086	0.03	0.43	889
	7	283					455	86		679	034	
		96					4					
38	-	-	0.105	-	0.149585	-0.271744	-	-	-	0.18	0.24	-
	0.45	0.5	736	0.070951			0.0	0.2	0.03	635	894	1.063
	91	348					455	583	679	3	2	55
		17					4					
39	0.04	-	1.850	-	-	-0.271744	0.3	0.2	0.12	-	1.77	0.517
	591	0.7	382	0.070951	0.149354		166	273	0086	0.03	733	889
		412					73	62		679	3	
		37										
40	1.05	0.0	-	0.447146	0.04642	-0.271744	-	0.2	0.18	-	-	3.249
	593	844	1.251				0.0	273	6353	0.16	0.09	466
		45	211				455	62		126	07	
							4					
41	0.55	-	-	0.04642	0.188687	-0.271744	-	-	0.12	-	0.75	0.014
	092	2.8	0.863				0.0	0.2	0086	0.03	840	703
		054	512				455	583		679	6	
		41					4					
42	2.23	-	0.105	0.488123	0.227362	-0.271744	0.3	0.2	-	-	0.58	-
	428	1.1	736				166	273	0.16	0.02	858	0.344
	7	540					73	62	126	634	5	72
		78										
43	1.72	0.4	0.493	0.174469	-	-0.271744	0.2	-	0.12	-	-	-
	927	972	435		0.149354		273	0.2	0086	0.03	0.26	0.057
	7	86					62	583		679	052	18
44	0.38	0.9	1.268	0.04642	0.413295	-0.271744	0.3	0.2	0.18	0.37	-	1.524
	258	101	833				166	273	6353	210	0.00	26
	3	26					73	62		6	579	

As no outliers were found in the data, all participants' data was retained for analysis.

3. **Sphericity:** A **Mauchly's Test of Sphericity** was conducted represent the results in Table 4.5 to assess whether the variances of differences between conditions (Zoom, Google Meet, and Prototype) were equal. The results of Mauchly's Test indicated that **sphericity was not violated** ($p = 0.1036$). This suggests that the assumption of **sphericity** is satisfied, and therefore no corrections (e.g., Greenhouse-Geisser or Huynh-Feldt) were needed for the repeated measures ANOVA.

TABLE 4.5: Mauchly's Test Results

Test	W (Test Statistic)	Chi-Square	Df	p-value
Mauchly's Test of Sphericity	0.8976	4.5354	2	0.1036

Given that all assumptions of normality, outliers, and sphericity were satisfied, the **repeated measures ANOVA** were conducted without the need for corrections.

4.8 Statistical Analysis

To determine whether the observed differences among Zoom, Google Meet, and the proposed prototype were statistically meaningful, a repeated measures analysis of variance (ANOVA) was conducted for each usability and workload metric. Since all participants interacted with all three systems, repeated measures ANOVA were considered appropriate for examining within-subject differences across conditions. This analysis was performed for effectiveness, efficiency, satisfaction, and cognitive workload.

TABLE 4.6: Repeated Measures ANOVA for Usability and Workload Metrics

Metric	F(2,86)	p-value	η^2	Result
Effectiveness	5.13	.008	0.106	Significant (Medium Effect)
Efficiency	1.87	.161	0.042	Not Significant (Small Effect)
Satisfaction	3.51	.034	0.075	Significant (Medium Effect)
Workload	0.04	.962	0.001	Not Significant

Table 4.6 presents the results of the repeated measures ANOVA for usability and workload metrics across the three systems, including F-values, p-values, and effect sizes, highlighting statistically significant differences. The ANOVA results revealed that there was a statistically

significant difference in **effectiveness** among the three systems, $F(2, 86) = 5.13$, $p = .008$, $\eta^2 = .106$. This indicates a statistically significant effect of system type on task effectiveness. Similarly, a statistically significant difference was observed for **user satisfaction**, $F(2, 86) = 3.51$, $p = .034$, $\eta^2 = .075$. The results reveal that users judged the ease of use of these systems in different ways, which means their satisfaction was also different in each system. On the other hand, there was no difference in terms of efficiency among these systems, $F(2, 86) = 1.87$, $p = .161$, $\eta^2 = .042$. Even though Google Meet was faster at completing tasks, this difference was not enough for being considered statistically significant. Likewise, there was no statistically significant difference regarding cognitive workload, $F(2, 86) = 0.04$, $p = .962$, $\eta^2 = .001$. This implies that differences in NASA-TLX scores among the three systems are very minor.

The values for the effect sizes support the analysis of the findings as well. First, the effectiveness measure had the effect size that was moderate. In other words, the difference between the systems made itself felt when it came to the success in completing the tasks. Meanwhile, satisfaction had a small but still significant effect size while both efficiency and workload were characterized by very small effect sizes. This suggests that the biggest distinctions between the systems concerned their ability to help users successfully complete tasks and be satisfied with using the systems.

4.8.1 Post-hoc Analysis

To compare systems more deeply, post-hoc comparisons were carried out using a Bonferroni correction. These analyses were done only on those variables where there was statistical significance found on the ANOVA test, specifically effectiveness and satisfaction.

TABLE 4.7: Post-hoc Pairwise Comparisons for Effectiveness (Bonferroni Adjusted)

Comparison	t-value	p (uncorrected)	p (Bonferroni)	Significance
Zoom vs Meet	-2.87	0.006	0.019	Significant
Zoom vs Prototype	-2.49	0.017	0.051	Marginal
Meet vs Prototype	0.43	0.667	1	Not Significant

Table 4.7 presents the Bonferroni-adjusted post-hoc pairwise comparisons for effectiveness across Zoom, Google Meet, and the proposed prototype. The results show values for t-statistics,

raw p-values, corrected p-values, and the level of significance for statistically significant differences in task success rates. Concerning the effectiveness of Zoom and Google Meet, there was a significant difference in their performances ($t = -2.87$, adjusted $p = .019$). These results mean that participants succeeded in performing tasks better using Google Meet rather than zoom. The comparison of Zoom and the prototype almost resulted in a significant difference in effectiveness ($t = -2.49$, adjusted $p = .051$). However, there was no statistically significant difference in the task success rate in terms of comparing Google Meet and the prototype ($t = 0.43$, adjusted $p = 1.00$).

TABLE 4.8: Post-hoc Pairwise Comparisons for Satisfaction (SUS) (Bonferroni Adjusted)

Comparison	t-value	p (uncorrected)	p (Bonferroni)	Significance
Zoom vs Meet	1.46	0.152	0.456	Not Significant
Zoom vs Prototype	-1.23	0.224	0.671	Not Significant
Meet vs Prototype	-2.61	0.012	0.037	Significant

Table 4.8 presents the Bonferroni-adjusted post-hoc pairwise comparisons for user satisfaction (SUS) across Zoom, Google Meet, and the proposed prototype. Accordingly, there are statistical differences in users' perceptions of usability between system pairings. As for the post-hoc analysis on user satisfaction, there was a statistically significant difference in mean ratings of user satisfaction between Google Meet and the prototype ($t = -2.61$, adjusted $p = .037$). This means that participants were more satisfied using the prototype than Google Meet. However, there were no statistically significant differences between Zoom and Google Meet ($t = 1.46$, adjusted $p = .456$) and between Zoom and the prototype ($t = -1.23$, adjusted $p = .671$). Overall, the most considerable improvement of user satisfaction was observed when comparing Google Meet with the prototype.

In terms of effectiveness, Google Meet was more effective than zoom, but there were no statistically significant differences between Google Meet and the prototype. In turn, the prototype was more satisfying for users compared to Google Meet, but other differences were not statistically significant. Hence, these results provide more detailed explanations of the overall ANOVA results since, in terms of effectiveness, the critical contrast was made between Zoom and Google Meet, whereas in terms of user satisfaction, the prototype prevailed over Google Meet.

Overall, the post-hoc analysis can be considered a more comprehensive interpretation of the significant ANOVA results since it allows specifying the essential contrasts in effectiveness and user satisfaction. Particularly, in terms of effectiveness, it is possible to state that there was a considerable difference between Zoom and Google Meet; as for satisfaction, the prototype was more beneficial for users than Google Meet. It means that, generally, the prototype was better from the perspective of usability.

These findings highlight that the assessment of usability cannot be performed based on objective measures alone because perceived usability and cognitive load must be considered alongside efficiency and effectiveness. Therefore, the prototype's ability to improve the instructor's experience was associated with the reduction of cognitive load and an increase in satisfaction levels. It indicates that it is important to implement HCI principles into SOLP development to facilitate instructors' performance in online settings.

4.9 Summary

The current chapter presented a thorough examination of the effectiveness, efficiency, and cognitive workload of three popular platforms for conducting synchronous learning online – namely Zoom, Google Meet, and the instructor-centered prototype suggested by this research. By considering both objective measures of performance as well as subjective users' satisfaction with the experience, the present work offered a complete evaluation of each platform's ability to promote efficient communication and task accomplishment during online synchronous teaching. Overall, Google Meet demonstrated higher levels of both effectiveness and efficiency by presenting the highest task success rate as well as a faster time-to-complete task compared to other platforms. However, although showing better performance in terms of effectiveness and efficiency, Google Meet yielded the lowest level of users' satisfaction with the experience. On the contrary, the instructor-centered platform showed high user satisfaction with interaction along with the lowest cognitive workload, which makes it more favorable in terms of usability in comparison to other two systems.

Further statistics also confirmed these findings in terms of effectiveness and user satisfaction; however, no differences in efficiency and cognitive workload were observed between the systems. Post-hoc analysis showed that Google Meet was found to be more effective than zoom; the instructor-centered system, in turn, yielded a statistically significant increase in user satisfaction when compared to Google Meet. As such, it was revealed that usability in synchronous learning systems is not only about effectiveness and efficiency. Moreover, the results of this research show how much the usability of such system can depend on the specific

design of the user interface and the level of associated cognitive workload.

In the conclusion of the present chapter, the performance of each system is summarized based on the mean values of all metrics used and presented through an ordered list.

TABLE 4.9: Mean Usability and Workload Scores by Tool

Metric	Zoom	Meet	Prototype
Effectiveness	87.88	92.27	91.67
Efficiency (Time) ↓	3.45	3.31	3.43
Satisfaction (SUS)	49.94	47.33	52.56
Cognitive Load ↓	31.16	30.89	30.66

The mean values of the most important usability and workload parameters for Zoom, Google Meet, and the suggested prototype are shown in Table 4.9. These findings provide a comprehensive overview of the system performance in terms of effectiveness, efficiency, user satisfaction, and workload.

TABLE 4.10: Final Ranking Summary of Systems Across Metrics

Metric	Best Tool	Why
Effectiveness	Meet	Highest mean + significant difference
Efficiency	Meet	Fastest (not significant)
Satisfaction	Prototype	Highest + significant
Cognitive Load	Prototype	Lowest (not significant)

Table 4.10 provides an analysis of the ranking of the systems based on the essential metrics of usability and workload. The ranking is based not only on the values but also on their levels of significance.

From the data presented in Table 4.8 and Table 4.9, Google Meet demonstrates good results in terms of effectiveness and efficiency while the prototype system obtains better results concerning user satisfaction and cognitive workload. The prototype has the best SUS value ($M = 52.56$), followed by Zoom ($M = 49.94$) and Google Meet ($M = 47.33$). However, all these values do not exceed the required level of 70 and therefore indicate low levels of usability for all systems.

Chapter 5

Conclusion And Future Work

5.1 Conclusion

The present study is devoted to the identification and overcoming of usability issues in synchronous online learning platforms (SOLPs). The current work focuses on Zoom and Google Meet as commonly used platforms and a novel SOLP prototype designed based on the Human–Computer Interaction (HCI) approach centered on instructors.

The empirical data revealed that both Zoom and Google Meet experience numerous usability challenges, such as difficult navigation, fragmented interface design, and high cognitive load. Thus, the results corroborate previous findings and demonstrate that the current platforms do not comply with usability requirements because they were originally developed for general communications purposes, not education.

To fill the gap, the research proposed an instructor-centered SOLP prototype based on ISO 9241-11 standards and Nielsen's usability heuristics. The prototype is characterized by task integration, clear navigation options, and better accessibility of control tools.

In this study, prototype usability was assessed through a within-subject experiment involving 44 participants. Both objective (task completion time) and subjective (System Usability Scale, SUS; NASA Task Load Index, NASA-TLX) measurements were gathered. The differences between the studied platforms were tested using Repeated Measures ANOVA.

It should be noted that the prototype outperformed other systems concerning various aspects of usability. First, its effectiveness was confirmed as the prototype demonstrated higher task completion rates compared to other platforms. As for efficiency, although the differences in task completion time did not reach significance, the prototype showed comparable results. Finally, the highest SUS score was observed for the prototype ($M = 52.56$), meaning that users perceive the system positively and consider it more convenient than the compared platforms. Nonetheless, SUS scores for all studied systems were lower than the desirable threshold of 70, indicating that improvements are necessary for increasing their usability.

As regards cognitive load, the NASA-TLX scores for the prototype were generally lower than those for other SOLPs, which means that users experienced less workload while interacting with the platform. This finding confirms the importance of developing platforms with user-

friendly interfaces as it allows instructors to concentrate fully on teaching.

Overall, the obtained results prove that an instructor-centered approach may be beneficial when designing SOLPs since it can significantly increase their usability.

5.2 Answers to Research Questions

The main research questions (RQ1, RQ2, and RQ3) were posed for addressing usability issues associated with synchronous online learning tools and possible advantages of the proposed instructor-focused prototype. Below, you will find the answers to each of the research questions posed.

- **RQ1: What usability problems arise in the use of synchronous online learning platforms (for example, Zoom and Google Meet) by instructors?**

The study brought out several usability problems experienced in both Zoom and Google Meet. Such usability problems as navigational challenges, lack of consistency in interface design, and increased cognitive load were among those that made it difficult for lecturers to engage in multiple activities simultaneously during class sessions. The problems included lack of consistent interfaces and error prevention tools, among others.

- **RQ2: Which usability principles and guidelines are violated in existing systems?**

Our analysis shows that both Zoom and Google Meet fail to meet several important usability guidelines. While Zoom violates the guidelines related to task appropriateness, error avoidance, and consistency, Google Meet fails to meet the guidelines associated with adaptability and learnability. These violations are summarized in Tables 3.1 and 3.2 in Chapter 3.

- **RQ3: Is it possible to achieve higher usability results by means of designing an instructor-centered prototype?**

According to the results, a prototype based on an instructor-centric approach shows improved usability. This can be seen from the high success rate in performing tasks, increased satisfaction among users, and decreased cognitive load compared to Zoom and Google Meet. While the prototype still falls below the 70 marks of SUS, it clearly performs better than the two competing software, thus proving the possibility of achieving improved usability through such an approach.

5.3 Hypotheses Testing

Hypothesis testing for usability improvements resulting from the suggested instructor-oriented prototype was conducted based on the established goals of the research. The results of the statistical analysis have led to the following conclusions:

- **H1: "There is no significant difference in the effectiveness of task completion between the instructor-centered prototype and existing platforms such as Zoom and Google Meet."**

Not Supported:

- **Post-hoc Analysis (Effectiveness):**
 - **Zoom vs Meet:** The comparison showed a significant difference ($p = 0.019$), with Meet performing better in terms of effectiveness.
 - **Zoom vs Prototype:** The comparison showed a marginal difference ($p = 0.051$), where the prototype had slightly lower effectiveness than zoom.
 - **Meet vs Prototype:** No significant difference ($p = 1.00$).
- The prototype did not significantly outperform Zoom or Meet in terms of effectiveness, with a marginal difference in comparison to Zoom represented in Table 4.7 Therefore, **H1 is not supported.**
- **H2: "There is no significant difference in the efficiency of task completion between the instructor-centered prototype and existing platforms such as Zoom and Google Meet."**

Not Supported:

- The time efficiency between the platforms, as shown in the main analysis table 4.9. However, the post-hoc pairwise comparisons indicate statistically significant differences in efficiency (p values for Zoom vs Meet, Zoom vs Prototype, and Meet vs Prototype all above the threshold for significance).
- **Conclusion: H2 is not supported**, as significant differences were found in the efficiency of task completion between the prototype and existing platforms.
- **H3: "The proposed prototype will achieve higher usability ratings compared to existing platforms."**

Supported:

- **Satisfaction (SUS) Analysis:**
 - **Zoom vs Meet:** No significant difference ($p = 0.456$).
 - **Zoom vs Prototype:** No significant difference ($p = 0.671$).
 - **Meet vs Prototype:** A significant difference ($p = 0.037$), with the prototype

achieving higher satisfaction ratings than Meet.

- The prototype achieved higher usability ratings, particularly when compared to Meet represented in Table 4.8. Hence, **H3 is supported**.
- **H4: "The proposed prototype will achieve lower cognitive workload compared to existing platforms."**

Supported:

- While the cognitive workload results are consistent with previous analysis, no post-hoc analysis was performed specifically on cognitive workload comparisons. However, the table 4.9 indicated that the prototype had the lowest cognitive load, as shown in the initial analysis, which supports this hypothesis.
- **Conclusion: H4 is supported**, as the prototype resulted in a lower cognitive workload compared to Zoom and Meet.

5.4 Summary of Contributions

Contribution of this work to the field of Human Computer Interaction is presented by means of providing an instructor-centered usability assessment of the synchronous online learning platform. Previous research in this area has mostly focused on students' experience or on general usability measures, while this paper presents a discussion of the difficulties of instructors' performance in this context.

One of the key contributions of the study is the combination of usability analysis and cognitive workload analysis into one experimental setting. The use of objective performance metrics along with standard measurements of user's perception based on SUS and NASA-TLX provides for a combined measure of the usability of the software product.

Another contribution made relates to the introduction of a highly realistic prototype from the instructor-centered viewpoint and its empirical evaluation. Unlike many other works in the field, which pay attention to separate usability problems or only conceptual designs, this study provides proof of improvements based on experimentation.

All in all, the results of the work make a considerable extension of current usability theory in terms of connection between interface design, task performance and cognitive workload.

5.5 Limitations of the Study

However, despite the above merits, the study is associated with several limitations which

should be considered during the interpretation of the results.

First, the total number of participants reached 44 people, which may limit the ability to generalize the findings. The within-subject experiment increases the validity of the study, but to draw conclusions applicable to a wide variety of cases, a sample consisting of a larger number of students from different educational settings would be needed.

Second, the study focuses only on a selected set of tasks characteristic of typical instruction performed by teachers in online learning environments. However, the complexity, dynamism, and variability of the actual classroom communication processes are likely to exceed the artificial tasks used in this experiment.

Third, the testing took place in an experimental setting, which means that the findings can only partly characterize the user behavior, adaptability of the application, and performance during the extended period of operation in a realistic environment.

Fourth, the evaluation of the prototype's performance was mainly based on standardized questionnaires, namely, the SUS and NASA-TLX scales. Despite their reliability and widespread application, adding qualitative techniques like interviewing, think aloud methods, or user discussion would provide additional insight into the use of the software.

Finally, the focus of the research was exclusively placed on instructor usability while disregarding the potential for assessing student-to-student interaction and collaboration. Therefore, there is plenty of room left for future studies concerning this issue. These limitations become apparent as the discussion unfolds below.

5.6 Future Work

Based on limitations and outcomes of this study, there are multiple opportunities for conducting future research in the field of synchronous e-learning usability.

First, it is highly recommended for researchers to include many participants to obtain more accurate information concerning the subject under consideration. In future studies, more instructors representing various disciplines and possessing experience in using different types of software should be involved. Thus, the obtained results will be more generalizable and useful.

Furthermore, future studies should combine evaluations provided by students and instructors since both these groups of people should use a particular software to make it efficient and

effective. Consequently, an analysis of usability for both these categories of participants will allow evaluating its overall quality.

Additionally, researchers can continue to conduct longitudinal usability studies to estimate learnability of the software prototype under development and interaction patterns of users that are rather difficult to observe during experimental sessions.

Furthermore, apart from quantitative data, it would be reasonable to conduct qualitative investigations that will include conducting interviews, usability walkthroughs, and think aloud protocols.

In addition, in future research, it might be possible to explore possibilities to use such advanced features as intelligent dashboards to provide necessary help and automate routine actions. Thus, instructors' cognitive load may significantly be reduced.

Moreover, researchers should consider expanding the proposed framework to cover other learning environments. For example, research conducted with hybrid and asynchronous learning environments might prove helpful.

5.7 Final Remarks

This research highlights the critical importance of usability in the efficiency of synchronous online learning systems. As online education evolves, it is becoming clear that the need for educational systems that meet the needs of usability as well as efficiency has become very important.

From the results obtained, it is apparent that while there may be many learning systems, they are far from meeting the usability requirements of the teachers. Designing systems based on the perspective of usability will bring about a lot of improvement in terms of usability as well as reduction in cognitive load.

It should be noted that the proposed model and prototype lay a good foundation for further studies in this area. This is one area where the findings of this study are expected to have an impact in the future.

Chapter A

Appendix A

Task Descriptions

The following tasks were designed to simulate typical teaching activities during an online class session. Participants were required to complete these tasks on each platform (Zoom, Google Meet, and the instructor-centered prototype). Each task was structured to assess the platform's usability and participants' ability to perform basic teaching-related actions.

1. **Login:** Log into the platform and verify successful access to the session.
2. **Join Meeting:** Join the meeting and ensure all required features are available (audio, video, etc.).
3. **View Camera:** Enable the camera and adjust settings to ensure proper visibility.
4. **Microphone Settings:** Adjust the microphone settings to ensure clear audio output.
5. **Use Shortcut Keys:** Use shortcut keys (if available) to perform specific actions (e.g., mute/unmute).
6. **View Class Activity & Material:** Share materials such as slides or videos with the class.
7. **Chat Usage:** Engage with the class by responding to questions in the chat.
8. **Minimize/Maximize Shared Screen:** Manage the shared screen, minimizing or maximizing it as needed.
9. **Start Recording:** Start recording the session to allow participants to access it later.
10. **Enable Captions:** Turn on captions for accessibility.
11. **Leave Class:** End the session for all participants and ensure proper class closure.
12. **Confirmation Before Leaving:** Verify that the platform prompts confirmation before leaving the class.
13. **Undo Leave Class:** Undo the action of leaving the class if done incorrectly.
14. **Report Issue Form:** Use the form or feature to report any issues encountered during the session.

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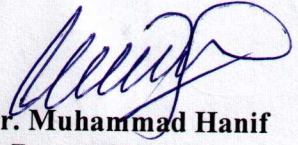
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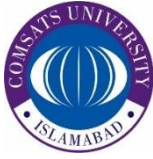

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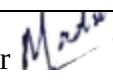
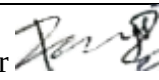
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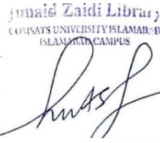
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